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Instrumental Variables Meet Machine Learning: Benchmarking GRF and DRIV for Heterogeneous Treatment Effects

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Abstract

This thesis benchmarks two machine learning methods, Doubly Robust IV (DRIV) and Generalized Random Forests (GRF), for estimating heterogeneous treatment effects under endogeneity. Using a controlled Monte Carlo simulation, we evaluate their finite-sample performance across varying sample sizes, instrument strengths and treatment effect complexities. The findings highlight that GRF outperforms DRIV in most settings, particularly under nonlinear heterogeneity and strong instrument compliance. GRF delivers low-bias estimates and adapts well to structural complexity, though it suffers from high variance in small samples with weak instruments. DRIV, by contrast, consistently exhibits structural bias and only performs competitively in large-sample linear settings. Notably, both methods are highly sensitive to local compliance: even when conventional first-stage F-statistics suggest strong instruments, weak local identification severely compromises estimator reliability. Under such conditions, both methods fail to outperform a classic Wald estimator. These results underscore that machine learning-based IV methods offer substantial advantages only when instruments are sufficiently strong and compliance is robust. Otherwise, their use may lead to misleading inference and poor policy recommendations.

1 Introduction

Estimating causal effects lies at the heart of empirical economics, particularly in domains such as labor, education and health, where policymakers aim to assess the impact of interventions. Traditionally, empirical work has focused on identifying average treatment effects (ATE) (Imbens and Wooldridge, 2009), which are often sufficient and efficient when treatment effects are homogeneous or when only average impacts matter. However, in many areas of applied economics, from education and labor markets to public health and taxation, the impact of an intervention can vary widely across individuals. In such settings, focusing solely on the ATE can obscure important heterogeneity that is crucial for policy targeting and understanding differential impacts (Athey and Imbens, 2017). Identifying which groups benefit most and under what conditions enables better targeting of limited resources and supports more equitable outcomes.

A further challenge in these contexts is that treatment assignment is often endogenous. Individuals may select into treatment based on unobserved traits such as motivation, preferences or private information, which also affect outcomes. For instance, in education, students with higher unobserved ability are both more likely to pursue further schooling and more likely to succeed regardless of treatment, complicating causal inference. Yet, the highest returns often go to those induced to study by policy, precisely the individuals policymakers aim to reach (Card, 2001). These patterns also appear in welfare, labor and tax policies, where individuals self-select into programs and the effects of treatment differ across subgroups. Accurately estimating heterogeneous effects under endogeneity is therefore essential for designing interventions that are both effective and equitable. In such settings, the relevant estimand is often the Conditional Local Average Treatment Effect (CLATE), which captures how treatment effects vary across compliers with different observed characteristics.

Instrumental variable (IV) techniques offer a way to recover causal effects under such conditions, but classical IV approaches typically estimate only average effects. For estimating heterogeneous effects, machine learning offers a potential solution to these problems. Over the past decade, machine learning methods have been increasingly adopted for causal inference, extending their role beyond prediction to include estimating treatment effects in flexible and high-dimensional settings (Lechner and Mareckova, 2025). Yet, most of these approaches rely on the assumption of unconfoundedness and thus fail under endogeneity. This creates a methodological tension: policymakers increasingly demand evidence on how effects vary across groups, yet the tools for estimating such heterogeneity under endogeneity remain underdeveloped and poorly understood in practice.

Recent advances in causal machine learning have begun to close this gap by extending IV methods to estimate Conditional Local Average Treatment Effects (CLATE). However, little is known about how these methods perform in finite samples when both endogeneity and treatment effect heterogeneity are present. This leaves applied researchers without clear guidance on when to prefer one estimator over another. This thesis addresses that gap by systematically evaluating the performance of machine learning-based IV estimators in recovering heterogeneous treatment effects under endogeneity. Using a Monte Carlo simulation framework, we compare the reliability of leading methods across a range of data conditions.

This thesis thus focuses on two estimators for Conditional Local Average Treatment Effects

(CLATE): generalized Random Forests (GRF; (Athey et al., 2019)) and Doubly Robust IV (DRIV; (Syrgkanis et al., 2019)), which is an extension of the Double Machine Learning (DML) framework (Chernozhukov et al., 2018). Both estimators are grounded in the instrumental variables framework and leverage machine learning to flexibly estimate heterogeneous effects while accounting for endogeneity. While both target the same estimand, they differ in how they balance flexibility and structure. GRF uses adaptive tree-based partitions to capture complex patterns in treatment effect heterogeneity. In contrast, DRIV uses a bias-corrected regression approach that projects the effect function onto a low-dimensional space, yielding more precise and efficient inference when heterogeneity is smooth or structured.

The focus on GRF and DRIV is motivated by Lechner and Mareckova (2025), who find that GRF and Double Machine Learning (DML) are the most promising methods for estimating heterogeneous treatment effects under unconfoundedness. This thesis builds on their findings by evaluating whether these methods also perform well in instrumental variable settings. While alternative CLATE estimators exist, they are either computationally intensive or less interpretable. In contrast, GRF and DRIV are supported by asymptotic theory, do not require substantial computational resources and are available in open-source software.

To evaluate the performance of GRF and DRIV in estimating CLATE under endogeneity, this thesis employs a controlled Monte Carlo simulation study. The simulation builds on Lechner and Mareckova (2025) by introducing a binary instrument and systematically varying three key dimensions: sample size, instrument strength and the shape of treatment effect heterogeneity. Because the true treatment effect is known by design, estimator performance can be evaluated precisely using bias, root mean squared error, mean absolute error and confidence interval coverage at the individual and population level.

The simulation results show that both DRIV and GRF perform poorly under weak compliance and require substantially stronger instruments than classical IV methods to accurately estimate heterogeneous effects. These limitations persist even in large samples, indicating that weak instruments fundamentally undermine the reliability of both methods. In contrast, when instruments are strong, both estimators demonstrate improved performance. GRF performs best overall and delivers the most accurate estimates in nonlinear settings, particularly for individual-level effects. DRIV suffers from persistent bias throughout all simulation environments. However, it achieves the lowest root mean squared error in linear settings with large samples, reflecting the efficiency gains from its parametric projection approach. For average effects among compliers, neither method consistently outperforms a classic Wald estimator. Nonetheless, GRF and DRIV are valuable for recovering heterogeneous treatment effects under endogeneity, provided instruments are strong and model assumptions match the data-generating process.

These findings highlight the need for systematic evaluation of machine learning methods in IV settings. This thesis addresses that need by providing the first direct simulation-based comparison of machine learning methods for heterogeneous treatment effects under endogeneity. By evaluating their performance across a range of realistic data-generating conditions, it offers practical guidance for applied researchers on the trade-offs and reliability of each method in finite samples. In addition, the thesis introduces a flexible and fully reproducible simulation framework that can be extended to assess other estimators or settings. This further supports transparency and replicability in causal machine learning research.

The remainder of the thesis is structured as follows. Section 2 reviews the literature on causal machine learning, instrumental variables and treatment effect heterogeneity. Section 3 introduces the formal identification framework and defines the key estimands. Section 4 details the methodology for GRF and DRIV. Section 5 describes the simulation design, including the data-generating process and performance metrics. Section 6 presents the results. Finally, Section 7 concludes and discusses directions for future research.

2 Literature Review

2.1 From ATE to CLATE

A foundational concern in empirical research is how to define and estimate the causal effect of a treatment or intervention. Whether studying a job training program, a new drug, or an educational policy, the key question is how outcomes differ between treated and untreated units. This idea is formalized in the potential outcomes framework (Rubin, 1974), which defines the effect of a treatment as the difference between the outcome a unit would have under treatment and the outcome under control. Since only one of these outcomes can be observed for each unit, researchers face the fundamental problem of causal inference (Rubin, 1978): we must infer causal effects by comparing units who are otherwise similar.

Randomized controlled trials (RCTs) ensure random treatment assignment, making the difference in outcomes an unbiased estimator of the Average Treatment Effect (ATE). However, in many economic and policy settings, RCTs are infeasible due to ethical, legal, or logistical concerns (Deaton and Cartwright, 2018). Researchers therefore turn to observational data, where treatment is not randomly assigned. In this context, identification typically relies on the assumption of unconfoundedness: that treatment assignment is as good as random given observed covariates. Unfortunately, this assumption often fails due to endogeneity from omitted variables, simultaneity, or reverse causality, leading to biased estimates (Angrist and Pischke, 2009). Alternative strategies are thus needed to recover credible causal effects.

An important solution to endogeneity in observational studies is the method of instrumental variables (IV), which identifies causal effects by leveraging exogenous variation in treatment induced by an instrument. A valid instrument influences treatment assignment but has no direct effect on the outcome, thereby mimicking random assignment under certain assumptions (Angrist and Pischke, 2009). When such an instrument is available, the IV estimand identifies the Local Average Treatment Effect (LATE), the average effect for compliers, a subpopulation whose treatment status is affected by the instrument (Imbens and Angrist, 1994). The economic motivation for this framework, as well as its empirical relevance across fields is discussed further in Section 2.2.

While IV methods address endogeneity and enable credible causal estimation, they typically recover average effects for specific subpopulations. However, growing evidence in applied economics shows that such averages may conceal important heterogeneity across individuals or subgroups (Athey and Imbens, 2017). Recognizing this, researchers have increasingly shifted from viewing heterogeneity as a nuisance to treating it as a central object of interest. As Imai and Ratkovic (2013) argue, understanding heterogeneous treatment effects is essential for detecting differential responses, optimizing treatment assignment and improving external validity. Imbens and Wooldridge (2009) document how modern methods, such as IV, difference-in-differences and regression discontinuity, are increasingly used to explore effect variation. Athey and Imbens (2017) describe this shift in focus as "precision policy," where policy design is guided by how effects vary across observable characteristics.

To formalize such variation, researchers estimate the Conditional Average Treatment Effect (CATE), which captures how treatment effects differ across covariate profiles. While earlier work by Hahn (1998) and Heckman et al. (1997) laid the foundation, the CATE has only recently been defined as a formal estimand. Abrevaya et al. (2015) provide a rigorous framework for CATE estimation, offering both theoretical underpinnings and practical motivation.

Although recent literature has emphasized the importance of treatment effect heterogeneity, most empirical and methodological work assumes unconfoundedness. As Wang et al. (2020) point out, even when researchers focus on heterogeneity rather than averages, this assumption often fails in observational settings and conventional CATE estimators can yield biased results with invalid inference, leading to misleading policy conclusions.

To address this, recent work has extended the instrumental variables framework to allow for heterogeneity, leading to the Conditional Local Average Treatment Effect (CLATE). CLATE generalizes the LATE by allowing treatment effects to vary with observed covariates. Frölich (2007) was among the first to formalize and estimate the CLATE using a flexible nonparametric IV approach, avoiding restrictive functional form assumptions. As Oprescu and Kallus (2024) note, CLATE is especially relevant in modern applications such as personalized medicine and algorithmic decision-making, where unobserved confounding is common and compliance behavior varies with observable characteristics.

This conceptual shift has been supported by both theoretical and empirical work. On the theoretical side, earlier contributions by Newey and Powell (2003) and Darolles et al. (2011) developed nonparametric and semiparametric IV frameworks to estimate heterogeneous relationships under endogeneity. Although these approaches do not define CLATE explicitly, they emphasize the econometric value of local effect estimation when treatment effects vary with covariates. Empirical studies are highlighted in Section 2.2.

2.2 Economic Motivation and Policy Relevance

Estimating heterogeneous treatment effects in the presence of endogeneity is of significant importance across various domains of applied economics. Many key policy questions depend not only on whether a particular intervention has an effect, but also on the distribution of these effects across individuals or groups. Understanding such variation is critical for the formulation, targeting and evaluation of public policy in fields such as education, labor, health and public finance.

One widely studied context is the estimation of returns to education. In this setting, endogeneity arises because individuals' schooling choices are influenced by unobserved characteristics such as ability or motivation, making simple correlations between education and earnings potentially biased. To overcome this, studies such as Angrist and Krueger (1991) and Card (2001) employ instrumental variable strategies that leverage variation in education induced by external factors. Card (2001) highlight that these effects are heterogeneous. In particular, individuals whose education decisions are altered by policy changes, such as compulsory schooling laws, often experience higher returns than the average individual. This finding has important implications for the design of educational policy, suggesting that resources may be more effectively allocated by targeting populations most responsive to intervention. In this context, the notion of conditional local average treatment effects becomes especially relevant, as it can also capture the causal effect for the marginal individual, who is typically the most affected by policy-induced changes in education. Ignoring such heterogeneity may lead to uniform education policies that overlook the distributional impacts and thereby fail to address inequality in educational outcomes. A parallel pattern appears in family-size research. Brinch et al. (2017) find that the causal effect of an additional child on older siblings' education varies in both magnitude and direction across households, with positive effects for some and negative effects for others. Their marginal treatment effect estimates highlight how average effects can conceal substantial heterogeneity, reinforcing the importance of accounting for such variation when designing more effective and equitable family policies.

Similar concerns arise in the evaluation of labor market and welfare programs. For example, Bitler et al. (2006) document that welfare reform in the United States produced noticeably different effects across the outcome distribution, with some individuals benefiting while others experienced adverse outcomes. This type of heterogeneity is consistent with theoretical labor supply models and has contributed to a shift in policy evaluation from aggregate outcomes to more distribution-sensitive assessments. In particular, analysing policy effects across the income distribution allows researchers to detect whether reforms disproportionately benefit or harm certain groups, which is of critical importance for equity considerations. In addition, many of these programs are affected by endogenous selection, as, for example, more optimistic individuals may be more likely to participate and also more likely to succeed regardless of the program.

Additional evidence comes from studies on health insurance expansions and tax policy, both of which are often subject to endogeneity concerns. For example, individuals' decisions to enroll in public health insurance or respond to tax incentives are typically influenced by unobserved preferences, information, or constraints. In the case of Medicaid expansions in the United States, Currie and Gruber (1996) and Cutler and Gruber (1996) demonstrate that these policies improved health outcomes for low-income and previously uninsured populations, while also inducing substitution away from private coverage. Similarly, in public finance, the impact of programs such as the Earned Income Tax Credit (EITC) and 401(k) retirement savings plans varies across individuals. Chetty et al. (2013) show that behavioral responses to the EITC depend critically on local knowledge about the policy's structure, implying that average treatment effects may mask important heterogeneity. Furthermore, Abadie (2003) address endogeneity in estimating the effect of 401(k) eligibility on personal savings and find heterogeneous effects.

Collectively, these contributions underscore the necessity of accounting for both endogeneity and treatment effect heterogeneity in empirical economic research. Beyond statistical accuracy, these considerations are essential for informing policies that aim to improve economic outcomes in a targeted manner.

2.3 Machine Learning Methods

Empirical research on causal effects increasingly involves high-dimensional covariates and complex confounding, where traditional econometric methods often struggle Knaus et al. (2018). Although machine learning (ML) was originally designed for prediction, recent advances have adapted it for causal inference by modifying algorithms to satisfy identification requirements.

Empirically, ML-based estimators have shown improved robustness and precision compared to classical approaches, particularly when accounting for many covariates (Baiardi and Naghi, 2024). These practical insights are supported by theoretical and simulation-based work. Under plausible conditions such as sparsity or smoothness, studies including Wager and Athey (2018), Farrell et al. (2021) and Colangelo and Lee (2020) show that causal ML achieves $\sqrt{n}$-consistency and valid inference. Together, these developments extend the scope of causal inference and offer applied researchers principled tools for estimation under challenging conditions.

Despite these advances, causal ML methods remain underused in empirical economics, even in settings where their advantages would be most pronounced (Lechner and Mareckova, 2025). As Lechner and Mareckova (2025) notes, ML's flexibility enables it to outperform classical approaches when heterogeneity is complex or unknown, making them particularly relevant for applied work seeking nuanced policy insights.

A prominent line of work in this area began with Causal Trees (Athey and Imbens, 2016), which use recursive partitioning to identify subgroups with different treatment effects. These were extended to Causal Forests by Wager and Athey (2018), ensemble methods that improve stability and allow for valid inference under unconfoundedness. Generalized Random Forests (GRF), developed by Athey et al. (2019), further advance this framework by solving local moment equations with adaptive partitioning tailored to the estimand of interest. GRF introduces a gradient-based splitting rule optimized for heterogeneity, offering formal asymptotic guarantees for both estimation and inference.

Empirical applications and simulations highlight the strength of these methods. Knaus et al. (2018) find that causal forests consistently rank among the top-performing methods across diverse data-generating processes. Similarly, Brantner et al. (2024) show that causal forests outperform parametric alternatives in detecting subtle patterns of heterogeneity across randomized trials, though they emphasize the need for large sample sizes.

In parallel, Double Machine Learning (DML), introduced by Chernozhukov et al. (2018), offers a flexible approach to causal estimation with high-dimensional controls by combining Neyman-orthogonal scores and sample splitting. This enables consistent estimation even when nuisance components are learned via regularized or nonparametric methods. Chernozhukov and Semenova (2021) extend this framework to heterogeneous effects, showing how the CATE can be estimated as best linear projections of orthogonalized signals. Applied implementations, such as Knaus et al. (2018), offer practical guidance for using DML to estimate heterogeneous effects in empirical settings.

An alternative line of work focuses on meta-learners, which approach CATE estimation by decomposing it into subproblems solvable with standard ML algorithms. Among these, the X-learner (Künzel et al., 2019) performs well under treatment imbalance, while simpler variants like the T- and S-learners tend to struggle in settings with complex heterogeneity (Credit and Lehnert, 2024).

A key limitation of many causal ML methods is their inability to address endogeneity. Recent advances overcome this by extending flexible estimators to the instrumental variables (IV) framework, enabling identification of Conditional Local Average Treatment Effects (CLATEs). Generalized Random Forests have also been extended to handle endogeneity through an instrumental variables framework. Athey et al. (2019) show that by solving localized IV moment conditions, GRF can be used to estimate Conditional Local Average Treatment Effects (CLATEs) without imposing strong functional form assumptions. This extension preserves GRF's nonparametric flexibility while allowing for valid inference. Applications such as Inoue et al. (2024) highlight its practical relevance for estimating heterogeneous effects in policy settings with non-random assignment.

Another key development is the Doubly Robust IV (DRIV) estimator by Syrgkanis et al. (2019), which extends the DML framework to accommodate both heterogeneity and endogeneity. DRIV constructs an initial estimate of the conditional treatment effect, then regresses a doubly robust transformation of the outcome on covariates. This approach leverages Neyman-orthogonal moments to reduce sensitivity to nuisance estimation while capturing treatment effect heterogeneity. Like IV GRF, it enables valid inference and performs well in high-dimensional settings.

Several other machine learning methods have been proposed for IV-based CATE estimation. DeepIV (Hartford et al., 2016) uses deep neural networks in a two-stage IV framework to flexibly model treatment and outcome processes, but poses significant computational and tuning challenges, making it less suitable for simulation-based comparisons. Alternative approaches, such as IV forests with refined splitting rules (Wang et al., 2020) and compliance-adjusted estimators (Oprescu and Kallus, 2024), aim to improve bias-variance trade-offs or account for heterogeneity in compliance.

Lechner and Mareckova (2025) conduct a comprehensive simulation study comparing GRF and DML (along with a modified causal forest) under unconfoundedness. They find that DML performs well for estimating average treatment effect or when heterogeneity is simple. GRF, by contrast, is better suited for capturing fine-grained heterogeneity (CATEs) due to its outcome-focused design and ability to model complex interactions. However, both methods have limitations: DML is sensitive to high selection bias, while GRF lacks internal consistency across aggregation levels. Yet, these methods have not yet been systematically compared in settings with endogenous treatment assignment. This thesis contributes to that effort by focusing on GRF and DML. While DML has been widely applied to average treatment effect estimation, its extension to heterogeneous effects in instrumental variable (IV) contexts remains relatively unexplored. Similarly, GRF is widely adopted for CATE estimation, but relatively unexplored in endogenous settings. This comparison is therefore both natural and timely. Alternative methods are either computationally intensive or relatively new.

3 Theoretical Framework

3.1 Potential Outcomes Framework

We begin by establishing the potential outcomes framework, which provides the foundation for defining and identifying causal effects (Rubin, 1974). Let $i \in \{1, \ldots, n\}$ index observational units. Each unit is characterized by a vector of pre-treatment covariates $\mathbf{X}_i \in \mathbb{R}^p$, a binary treatment indicator $T_i \in \{0, 1\}$ and an observed outcome $Y_i^{\text{obs}} \in \mathbb{R}$. The treatment variable equals one if the unit receives the treatment and zero otherwise. Covariates are assumed to be pre-treatment and unaffected by assignment.

Under the potential outcomes framework, each unit has two well-defined potential outcomes: $Y_i(1)$, the outcome under treatment and $Y_i(0)$, the outcome under control. The individual treatment effect is defined as:

$$\theta_i = Y_i(1) - Y_i(0).$$

The unobserved potential outcome means that for each unit we see only one of $\{Y_i(1), Y_i(0)\}$, so θ_i cannot be observed directly. This is the fundamental problem of causal inference (Rubin, 1978).

To make $Y_i(0)$ and $Y_i(1)$ well-defined, we assume the Stable Unit Treatment Value Assumption (SUTVA) (Rubin, 2005), which requires:

1. No interference: one unit's treatment cannot affect another unit's outcomes.
2. No hidden versions: there is a single, consistent definition of treatment and of control.

3.2 Treatment Effect Estimands

We now define several causal estimands of interest within the potential outcomes framework.

3.2.1 Average Treatment Effect (ATE)

The most basic population-level estimand is the Average Treatment Effect (ATE), defined as:

$$\text{ATE} := \mathbb{E}[Y(1) - Y(0)].$$

It represents the average causal effect of the treatment across the entire population. While widely used, the ATE assumes homogeneity in treatment effects or is interpreted as an average over potentially heterogeneous effects.

3.2.2 Conditional Average Treatment Effect (CATE)

When heterogeneity in treatment effects is expected or of substantive interest, the Conditional Average Treatment Effect (CATE) becomes the primary estimand of interest:

$$\text{CATE}(x) := \mathbb{E}[Y(1) - Y(0) \mid \mathbf{X} = x].$$

This function-valued parameter characterizes how the treatment effect varies with observed characteristics, enabling subgroup or individual-level policy analysis and is central to modern approaches in personalized decision-making and precision treatment allocation. However, when the treatment is endogenous, the CATE is generally not point-identified without further assumptions or instruments.

3.2.3 Local Average Treatment Effect (LATE)

When using instrumental variables, we do not recover the Average Treatment Effect (ATE) for the full population. Instead, IV identifies the Local Average Treatment Effect (LATE), the average causal effect of treatment for a specific subpopulation known as compliers. Compliers are defined as those individuals whose treatment status is influenced by the instrument: they receive the treatment if encouraged by the instrument and do not receive it otherwise (Imbens and Angrist, 1994).

Formally, assuming a binary instrument $Z \in \{0, 1\}$ and a binary treatment $T \in \{0, 1\}$, the LATE is identified as:

$$\text{LATE} := \mathbb{E}[Y(1) - Y(0) \mid \text{Compliers}],$$

where the complier group consists of units for whom $T_i(1) = 1$ and $T_i(0) = 0$. This group is not directly observed but is defined implicitly by the monotonicity assumption, which rules out defiers (See section 3.3 for details) (Angrist et al., 1996).

3.2.4 Conditional Local Average Treatment Effect (CLATE)

To combine the heterogeneity of CATE with the identification strategy of LATE, the Conditional LATE (CLATE) (Frölich, 2007) is defined as:

$$\text{CLATE}(x) := \mathbb{E}[Y(1) - Y(0) \mid \mathbf{X} = x, \text{Compliers}].$$

This estimand captures the average treatment effect for compliers, conditional on covariates. In doing so, the CLATE allows for policy-relevant causal statements that are heterogeneous and can be identified with endogenous data.The CLATE is the central estimand of interest in this thesis.

3.3 Identification Assumptions

Let $Z \in \{0, 1\}$ be a binary instrument. Following Frölich (2007), identification of the Conditional Local Average Treatment Effect (CLATE) relies on five conditions. First, we have the relevance condition. This states that the instrument must affect the probability of receiving treatment:

$$\mathbb{P}(T = 1 \mid Z = 1, \mathbf{X} = x) \neq \mathbb{P}(T = 1 \mid Z = 0, \mathbf{X} = x).$$

This condition is testable in the data through a strong first-stage regression.

Second, we have the exclusion restriction, stating that the instrument affects the outcome only through its impact on the treatment:

$$Y(z, t) = Y(t).$$

This assumption is not testable and must be justified on theoretical grounds; see Angrist et al. (1996) for discussion.

Third, the independence assumption assumes that, conditional on covariates, the instrument is independent of potential outcomes and potential treatments:

$$Z \perp\!\!\!\perp (Y(0), Y(1), T(0), T(1)) \mid \mathbf{X}.$$

This assumption is plausible in settings such as randomized encouragement designs or natural experiments, although it cannot be verified empirically.

Fourth, we must have monotonicity, meaning that the instrument can only increase treatment uptake:

$$T_i(1) \geq T_i(0) \quad \text{for all } i.$$

That is, there are no individuals who would take the treatment when not encouraged but not take it when encouraged. This guarantees that compliers form a well-defined subpopulation (Angrist et al., 1996).

Fifth and last, the overlap assumption means that both values of the instrument must occur for all covariate values:

$$0 < \mathbb{P}(Z = 1 \mid \mathbf{X} = x) < 1.$$

Under these conditions, the Conditional Local Average Treatment Effect is identified.

4 Methodology

4.1 Double Machine Learning (DML)

4.1.1 Generic DML Framework

Recent advances in causal machine learning have enabled the estimation of treatment effects in high-dimensional and complex settings. However, naively applying machine learning (ML) to estimate structural parameters often yields biased and unreliable results. The Double Machine Learning (DML) framework, introduced by Chernozhukov et al. (2018), addresses this issue by constructing estimators that are robust to errors in nuisance function estimation.

Throughout, let $\{W_i\}_{i=1}^n$ denote an i.i.d. sample with elements $W_i = (Y_i, T_i, X_i, Z_i)$, where Y is the outcome, T the binary treatment, Z the binary instrument and $X \in \mathbb{R}^{d_X}$ a vector of controls, with d_X the dimension of X.

Suppose the causal parameter of interest $\theta_0 \in \Theta$ is characterised by a population moment equation

$$\mathbb{E}\big[\, m\big(W, \theta_0, \eta_0\big)\,\big] \;=\; 0, \tag{1}$$

where $\eta_0 \in \mathcal{T}$ collects nuisance functions such as propensity scores, outcome models, or conditional densities. In a classical low-dimensional setting one would insert estimators of η_0 directly into 1 and solve for θ_0.

However, when the nuisance parameter space $\mathcal{T}$ is high-dimensional, a naive plug-in estimator

$\tilde{\theta}$ obtained by solving

$$\frac{1}{n}\sum_{i=1}^{n} m\big(W_i, \tilde{\theta}, \tilde{\eta}\big) = 0$$

incurs two non-negligible biases. First, regularization bias arises because the penalties or shrinkage inherent to high-dimensional estimation systematically pull $\tilde{\eta}$ away from η_0, causing a persistent error in the moment condition. Second, overfitting bias emerges when the same data are used both to fit $\tilde{\eta}$ and to evaluate the empirical moments, inducing a spurious correlation between the estimation error $\tilde{\eta} - \eta_0$ and the score $m(W; \theta_0, \eta_0)$. These biases can remain substantial even in large samples. In contrast, sampling variation typically decreases at the usual $\sqrt{n}$ rate. Because regularization and overfitting biases are often much larger than this, the naive estimator $\tilde{\theta}$ may fail to be consistent or asymptotically normal, making standard inference invalid (Ahrens et al., 2025).

To combat regularisation bias, Chernozhukov et al. (2018) propose choosing the score in 1 to be Neyman orthogonal. This ensures that small errors in the nuisance inputs affect the moment condition only to second order. Formally,

$$\frac{\partial}{\partial \lambda}\, \mathbb{E}\big[\psi\big(W, \theta_0, \eta_0 + \lambda(\eta - \eta_0)\big)\big]\Big|_{\lambda=0} \;=\; 0 \quad \text{for all } \eta \in \mathcal{T}. \tag{2}$$

In simpler terms, this condition ensures that small mistakes in estimating the nuisance functions η_0 do not have a large impact on the moment condition used to estimate θ_0. As emphasised by Ahrens et al. (2025), this property allows researchers to use flexible, high-dimensional ML methods for nuisance estimation without compromising valid inference for the low-dimensional target θ_0.

Overfitting bias is mitigated through cross-fitting, a form of sample splitting that prevents the same observations from influencing both nuisance estimation and moment evaluation. Specifically, divide the data into K folds $\mathcal{I}_1, \ldots, \mathcal{I}_K$. For each fold k, estimate the nuisance functions $\widehat{\eta}_{0,-k}$ using only the data in the complementary set $\mathcal{I}_{-k} = \{1, \ldots, n\} \setminus \mathcal{I}_k$ and evaluate the score function on the held-out fold $\mathcal{I}_k$: $\psi\big(W_i, \theta, \widehat{\eta}_{0,-k}\big)$ for $i \in \mathcal{I}_k$. Because W_i is not used in constructing $\widehat{\eta}_{0,-k}$, the resulting empirical moments are conditionally unbiased. The final DML estimator is then obtained by averaging the cross-fitted score contributions across all folds:

$$\frac{1}{n}\sum_{k=1}^{K}\sum_{i \in \mathcal{I}_k} \psi(W_i, \widehat{\theta}, \widehat{\eta}_{0,-k}) = 0,$$

Under mild regularity and rate conditions on the ML learners, the DML algorithm delivers an asymptotically normal, root-n consistent estimator:

$$\sqrt{n}(\widehat{\theta} - \theta_0) \;\xrightarrow{d}\; \mathcal{N}(0, V), \qquad V = J^{-1}\,\mathbb{E}\Big[\psi(W, \theta_0, \eta_0)\psi(W, \theta_0, \eta_0)^{\top}\Big]\; J^{-1},$$

where $J = \mathbb{E}\left[\partial_{\theta}\psi(W, \theta_0, \eta_0)\right]$ (Chernozhukov et al., 2018).

Thanks to the Neyman-orthogonality of the score and cross-fitting, it is sufficient that the nuisance estimators converge at any rate faster than $n^{-1/4}$. Without orthogonality, one can still obtain consistency, but root-n asymptotic normality would generally require much faster convergence rates for the nuisance estimators and finite-sample bias is typically larger.

Appendix B demonstrates empirically that the combination of Neyman orthogonal scores and cross-fitting substantially mitigates bias and improves the validity of inference.

4.1.2 From generic DML to IV-CATE

The standard DML framework assumes unconfoundedness and primarily targets average treatment effects (ATE), not heterogeneity. Syrgkanis et al. (2019) extend the Double Machine Learning (DML) framework to settings that involve both endogeneity and heterogeneous treatment effects, enabling the estimation of conditional local average treatment effects $\theta_0(X)$. Their approach has two stages: the DMLIV estimator provides a preliminary, partially debiased estimate of $\theta_0(X)$. The second-stage DRIV estimator corrects remaining bias and yields the final reported estimates.

Stage 0: a preliminary CLATE via DMLIV

We start from the heterogeneous linear IV model

$$Y \;=\; \theta_0(X)\,T \;+\; f_0(X) \;+\; e, \qquad \mathbb{E}[e \,|\, Z, X] = 0, \tag{3}$$

where Y is the outcome, T the potentially endogenous treatment and e an error term orthogonal to the instrument Z conditional on X.

To convert the IV problem into an orthogonal moment, Syrgkanis et al. (2019) first residualise both the outcome and the treatment with respect to X. Let

$$q_0(X) = \mathbb{E}[Y \,|\, X], \qquad p_0(X) = \mathbb{E}[T \,|\, X], \qquad h_0(Z, X) = \mathbb{E}[T \,|\, Z, X].$$

Given estimators $\widehat{q}, \widehat{p}, \widehat{h}$, define the residualised variables

$$\widetilde{Y} = Y - \widehat{q}(X), \qquad \widetilde{T} = \widehat{h}(Z, X) - \widehat{p}(X).$$

Here, $\widetilde{T}$ is the part of the treatment that is induced by the instrument after the direct dependence on X has been partialled out. Regressing $\widetilde{Y}$ on $\widetilde{T}$ with a flexible learner (using cross-fitting as in Section 4.1.1) yields the pilot estimator $\widehat{\theta}_{\text{pre}}(X)$. The loss function is Neyman-orthogonal with respect to q_0 and p_0, so slow-rate estimation errors in those nuisance functions do not affect the consistency of $\widehat{\theta}_{\text{pre}}(X)$. However, the loss is not orthogonal with respect to h_0, so remaining bias may still arise from $\widehat{h}$. Formal proofs are provided in Appendix C.1.

Proper identification relies on the instrument strength measured in $\widetilde{T}$, which has conditional variance

$$V_0(X) \;=\; \mathbb{E}\big[(h_0(Z, X) - p_0(X))^2 \;\big|\; X\big] \;=\; \mathrm{Var}\big(\mathbb{E}[T \mid Z, X] \mid X\big). \tag{4}$$

When the instrument is weak for a given covariate value X, the residualised treatment $\widetilde{T}$ (and hence $V_0(X)$) shrinks towards zero. The local estimating equation for $\theta_0(X)$ divides by $V_0(X)$, meaning an arbitrarily small variance would make the parameter inflate its sampling error.

Thus, to properly identify $\widehat{\theta}_{\text{pre}}(X)$, we must have that

$$V_0(X) \;\geq\; \lambda \;>\; 0 \qquad \text{for all } X. \tag{5}$$

This overlap condition ensures the instrument induces enough within-X variation to identify the pilot estimator. It is the heterogeneous analogue of the relevance condition in Section 3.3 and is closely connected to the compliance score introduced in Section 5.2.3. Although $\lambda > 0$ is required, no specific value is prescribed, leaving diagnostics to the practitioner.

Stage 1: Doubly Robust IV estimator (DRIV)

While $\widehat{\theta}_{\text{pre}}(X)$ is orthogonal to errors in the outcome and propensity models, it remains sensitive to estimation errors of the first-stage function $h_0(Z, X)$. To address this, Syrgkanis et al. (2019) propose a second-stage debiasing step called Doubly Robust IV (DRIV), which constructs a pseudo-outcome regression based on a loss function that is orthogonal to all nuisance components.

First, let

$$r_0(X) = \mathbb{E}[Z \mid X], \qquad f_0(X) = \mathbb{E}[T \cdot Z \mid X]$$

.

Split the sample once into two folds S_1, S_2[1]. All nuisance functions $\widehat{q}, \widehat{p}, \widehat{r}, \widehat{f}$ and the pilot estimate $\widehat{\theta}_{\text{pre}}$ are trained on S_1; then, for each observation in the hold-out fold S_2, compute the cross-fit residuals

$$\widetilde{Y} = Y - \widehat{q}(X), \qquad \widetilde{T} = T - \widehat{p}(X), \qquad \widetilde{Z} = Z - \widehat{r}(X).$$

.

Define the functions

$$\beta_0(X) := \mathbb{E}\big[\widetilde{T}\,\widetilde{Z} \mid X\big], \qquad \widehat{\beta}(X) = \widehat{f}(X) - \widehat{p}(X)\,\widehat{r}(X),$$

where $\widehat{f}$ is obtained by regressing $T \cdot Z$ on X using the training fold S_1.

Given the pilot CLATE $\widehat{\theta}_{\text{pre}}(X)$, DRIV constructs the doubly robust score

$$Y^{\text{DR}} \;=\; \widehat{\theta}_{\text{pre}}(X) + \frac{\big(\widetilde{Y} - \widehat{\theta}_{\text{pre}}(X)\,\widetilde{T}\big)\,\widetilde{Z}}{\widehat{\beta}(X)}. \tag{6}$$

This expression is the Neyman-orthogonal score derived in Syrgkanis et al. (2019). Intuitively, when $\widehat{\theta}_{\text{pre}}$ is accurate the correction term vanishes, whereas large deviations are compensated by the residualised Wald ratio on the right. The final estimate $\widehat{\theta}_{\text{DR}}(X)$ is obtained by regressing Y^{DR} on X using any square-loss learner on fold S_2. Swapping folds and averaging completes the two-fold cross-fit.

[1] For simplicity and computational efficiency, we follow the original presentation in Syrgkanis et al. (2019) and use a single two-fold cross-fitting scheme throughout. However, this restriction is without loss of generality: the procedure can be extended to K-fold cross-fitting exactly as described in Section 4.1.1, with the same asymptotic guarantees.

The DRIV estimator can therefore be characterised as the minimiser of the quadratic risk

$$L_2(\theta) = \mathbb{E}\Big[\big(Y^{\mathrm{DR}} - \theta(X)\big)^2\Big], \tag{7}$$

optionally restricted to a hypothesis space Θ_π. Because Y^{DR} is Neyman-orthogonal, L_2 is insensitive to first-order errors in the nuisance estimates $\theta_{\mathrm{pre}}, \beta, q, p, r$. A formal proof of this orthogonality property is provided in Appendix C.2. Conditional on the nuisances, expression (7) is an ordinary least-squares objective, so any regression algorithm that minimises it yields valid $\sqrt{n}$ inference provided the nuisance learners converge at rate $o(n^{-1/4})$.

A key regularity requirement for the second stage is a compliance lower bound, $\beta_0(X) \geq \beta_{\min} > 0$. Similar to the overlap condition in (5) and the compliance score discussed in 5.2.3, this assumption rules out covariate regions where the instrument causes little or no change in the treatment. If $\beta_0(X)$ were allowed to approach zero, the denominator in (6) would inflate finite–sample variance and destroy the convergence properties. Again, the authors do not specify a particular value of $\beta_{\min}$.

Under these boundedness assumptions and assuming the nuisance estimators converge at rate $o(n^{-1/4})$, the DRIV estimator achieves parametric (root-n) convergence rates and is asymptotically normal:

$$\sqrt{n}\big(\widehat{\theta}_{\mathrm{DR}} - \theta^*\big) \xrightarrow{d} \mathcal{N}(0, \Sigma),$$

where Σ is the efficient variance (See Corollaries 4 and 6 in Syrgkanis et al. (2019)).

When the CLATE is projected onto a low-dimensional linear model, the orthogonality of the DRIV score ensures that standard OLS variance formulas remain valid, even though the nuisance functions are estimated by any machine learner. This allows valid inference for the projected coefficients using conventional methods (Syrgkanis et al., 2019).

It is important to emphasize that the DRIV estimator does not seek to recover the fully nonparametric function $\theta_0(x)$ directly. Instead, it estimates its projection onto a restricted hypothesis class Θ_π, such that

$$\theta^* = \arg\min_{\theta \in \Theta_\pi} \mathbb{E}[(\theta_0(X) - \theta(X))^2].$$

This yields the best L^2 approximation to $\theta_0(x)$ within Θ_π. In this thesis, we adopt the most generally used linear specification (Syrgkanis et al. (2019)), where Θ_π consists of linear functions over basis features $\phi(X)$, providing a transparent and interpretable summary of treatment effect heterogeneity.

In contrast, fully nonparametric methods like Generalized Random Forests (GRF) estimate $\theta_0(x)$ directly, allowing for greater flexibility but often at the cost of higher variance, reduced interpretability and increased computational complexity. DRIV thus offers a bias-variance trade-off through the choice of Θ_π, balancing accuracy, interpretability and efficiency.

4.2 Generalized Random Forests (GRF)

4.2.1 Generic GRF Framework

DML methods like DRIV estimate treatment heterogeneity using global models with fixed functional forms. Generalized Random Forests (GRF) overcome this by learning localized neighborhoods around each target point through adaptive tree-based partitioning. GRF thus combines nonparametric flexibility with valid inference via orthogonal estimating equations (Athey et al., 2019).

Let $\{W_i\}_{i=1}^n$ be an i.i.d. sample with $W_i = (Y_i, T_i, X_i, Z_i)$. The parameter of interest $\theta(x)$ and nuisance $\eta(x)$ are defined through the local moment condition:

$$\mathbb{E}\big[\psi_{\theta(x),\eta(x)}(W_i) \mid X_i = x\big] = 0.$$

GRF approximates this by solving a weighted version:

$$(\hat{\theta}(x),\, \hat{\eta}(x)) \in \arg\min_{\theta,\eta} \left\{ \left\| \sum_{i=1}^{n} \alpha_i(x)\, \psi_{\theta,\eta}(W_i) \right\|_2^2 \right\}.$$

where weights $\alpha_i(x)$ define neighbor relevance around each target x.

Each tree partitions the feature space recursively. For query point x, define its leaf in tree b as $L_b(x)$, the set of training points sharing the same split path. The per-tree weights are

$$\alpha_{b,i}(x) = \frac{\mathbf{1}\{X_i \in L_b(x)\}}{|L_b(x)|}, \quad \alpha_i(x) = \frac{1}{B}\sum_{b=1}^{B} \alpha_{b,i}(x).$$

Intuitively, GRF automatically learns which observations are "neighbors" of x by building these trees, rather than fixing a single radius or bandwidth. Because the splits are chosen to expose variation in the moment conditions we care about, this adaptive neighborhood will concentrate on the most informative data around x.

Tree splits are chosen to target heterogeneity. Given a parent node P with children C_1, C_2, the split quality is measured by the Δ-criterion

$$\Delta(C_1, C_2) := \frac{n_{C_1} n_{C_2}}{n_P^2} \left(\hat{\theta}_{C_1} - \hat{\theta}_{C_2}\right)^2,$$

where $\hat{\theta}_{C_j}$ is the local moment estimate in child C_j. This criterion favors splits that create child nodes with both sufficient size and pronounced differences in the estimated effect. As a result, the tree recursively partitions the covariate space in a way that adapts to the underlying heterogeneity in $\theta(x)$.

Since full computation is costly, Athey et al. (2019) propose a gradient-based approximation using pseudo-outcomes:

$$\rho_i = -\xi^\top A_P^{-1} \psi_{\hat{\theta}_P, \hat{\eta}_P}(W_i),$$

with A_P the Jacobian of the score and ξ a user-defined direction. Regression trees then split on ρ_i, efficiently approximating the Δ-criterion.

To avoid bias, GRF employs honesty: for each tree, the data is split into a structure half for splitting and an estimation half for computing local moments. This ensures valid asymptotic inference, similar to cross-fitting in DML (Wager and Athey, 2018).

To put it all together, each tree in the forest is trained on a subsample of the original data. Within this subsample, observations are randomly split into a structure half and an estimation half. The tree is grown on the structure half by recursively partitioning the covariate space to maximize the gradient-based splitting criterion. Once the tree is fully constructed, the estimation half is used to compute local estimates of the target parameter in each leaf by solving the sample moment conditions. Repeating this across all B trees yields, for any target point x, a set of forest weights $\{\alpha_i(x)\}_{i=1}^n$ defined by averaging leaf memberships. The final estimate $\hat{\theta}(x)$ is then obtained by solving the locally weighted moment equation using these weights.

Inference follows Athey et al. (2019). Honest splitting ensures asymptotic normality. Variance is estimated via the bootstrap of little bags: trees are grouped into bags sharing subsamples and an ANOVA-style decomposition captures score variability (Sexton and Laake, 2009). Combined with a delta-method correction, the standard error at point x is

$$\mathrm{SE}(x) = \sqrt{\hat{\sigma}^2(x)}, \quad \text{with} \quad \hat{\theta}(x) \pm z_{1-\alpha/2}\, \mathrm{SE}(x)$$

yielding a valid confidence interval. Theorem 6 in Athey et al. (2019) establishes consistency and asymptotic validity under regularity conditions. For theoretical details on the derivation of the standard errors, see Appendix D.

4.2.2 Application to Instrumental Variables

We adapt the GRF framework to the IV setting by solving local moment equations that identify $\theta(x)$ under standard IV assumptions. Each observation is given by $W_i = (Y_i, T_i, Z_i, X_i)$, where Z_i is a valid instrument for the endogenous treatment T_i and $X_i \in \mathbb{R}^{d_X}$ are covariates.

At each target point x, GRF estimates $(\theta(x), \mu(x))$ by solving:

$$\begin{aligned} \mathbb{E}\left[Z_i \cdot (Y_i - \mu(x) - \theta(x)T_i) \mid X_i = x\right] &= 0, \\ \mathbb{E}\left[Y_i - \mu(x) - \theta(x)T_i \mid X_i = x\right] &= 0, \end{aligned}$$

where $\mu(x)$ is a local intercept. The first condition ensures identification of $\theta(x)$, while the second centers residuals to stabilize estimation.

Replacing expectations with forest-weighted sample analogues, the estimator $(\hat{\theta}(x), \hat{\mu}(x))$ solves:

$$\sum_{i=1}^{n} \alpha_i(x) \cdot \begin{bmatrix} Z_i \cdot (Y_i - \mu - \theta T_i) \\ Y_i - \mu - \theta T_i \end{bmatrix} = 0.$$

This is equivalent to a locally weighted IV regression:

$$\theta(x) = \frac{\mathrm{Cov}(Y_i,\, Z_i \mid X_i = x)}{\mathrm{Cov}(T_i,\, Z_i \mid X_i = x)}.$$

To guide splitting toward heterogeneous treatment effects, Athey et al. (2019) propose a gradient-based pseudo-outcome. For a parent node P, define:

$$\rho_i = (Z_i - \bar{Z}_P) \cdot \left[(Y_i - \bar{Y}_P) - \hat{\theta}_P (T_i - \bar{T}_P) \right],$$

where bars denote within-node means and $\hat{\theta}_P$ is the 2SLS estimate on node P. This pseudo-outcome approximates the gradient of the moment condition with respect to θ. Tree structure is learned on these ρ_i, while parameter estimation is done on a separate subsample to ensure honesty.

To further improve finite-sample performance and reduce bias, it is beneficial to orthogonalize variables prior to estimation:

$$\widetilde{Y}_i = Y_i - \widehat{\mathbb{E}}[Y_i \mid X_i], \quad \widetilde{T}_i = T_i - \widehat{\mathbb{E}}[T_i \mid X_i], \quad \widetilde{Z}_i = Z_i - \widehat{\mathbb{E}}[Z_i \mid X_i],$$

where conditional expectations are estimated using separate regression forests on out-of-bag samples. The moment conditions are then applied to the residualized variables.

The final IV-GRF estimator $\hat{\theta}(x)$ is obtained by solving a locally weighted IV problem using forest weights and optionally orthogonalized inputs. Under regularity conditions, the estimator is consistent and asymptotically normal. Inference proceeds via the bootstrap of little bags, as described in the previous subsection.

4.3 Benchmark: Wald Estimator

To contextualize the performance of modern estimators for heterogeneous treatment effects under endogeneity, we include the classical Wald estimator as a benchmark throughout the simulation study. Originally introduced by Wald (1940), the Wald estimator is a simple and widely-used instrumental variables estimator that recovers the Local Average Treatment Effect (LATE) under standard IV assumptions. For a binary instrument $Z \in \{0, 1\}$, the Wald estimator is given by

$$\hat{\theta}_{\text{Wald}} = \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[T|Z=1] - \mathbb{E}[T|Z=0]}, \tag{8}$$

which corresponds to the ratio of reduced-form to first-stage differences in means. This estimator is a special case of the IV estimator when both treatment and instrument are binary and is interpreted as the effect of treatment on the treated compliers (Angrist and Pischke, 2009).

Although the Wald estimator is inherently limited to identifying average effects among compliers and does not account for heterogeneity, it remains a valuable reference point for evaluating more flexible estimators like DRIV and GRF. In particular, if either method fails to outperform the Wald estimator, despite specifically targeting the Conditional Local Average Treatment Effect (CLATE), this would call into question the practical usefulness of their added complexity. In CLATE estimation, the Wald estimator effectively yields a constant prediction for all individuals. Thus, if GRF or DRIV cannot improve upon this naive benchmark even when heterogeneity is present, they may fail to capture meaningful individual-level variation. Including the Wald estimator therefore provides a lower bound on performance and ensures that our simulation results can be interpreted relative to a standard IV estimator.

5 Simulation design

5.1 Overview and Motivation

The Monte Carlo simulation study is designed to compare the finite-sample performance of the DRIV method and the GRF method. Both methods are evaluated under a controlled endogeneity setting in which treatment assignment depends on unobserved confounders and a valid binary instrument is required to recover causal effects. The simulation setup builds on the framework of Lechner and Mareckova (2025), adopting key elements such as the data-generating process for covariates, outcome structures and treatment effect heterogeneity. This ensures methodological consistency with prior work. The main extension is the introduction of a binary instrument into the latent treatment index, creating endogeneity and enabling analysis of estimator performance under varying levels of instrument strength. To explore estimator behavior across realistic settings, we vary sample size, treatment effect complexity and instrument strength.

5.2 Data-Generating Process

5.2.1 Covariates

Following the structure proposed by Lechner and Mareckova (2025), the simulated data consist of $p = 10$ covariates, with $u = 5$ drawn independently from a standardized uniform distribution and $n = 5$ from a standard normal distribution. Of these, only the first $k = 6$ covariates directly influence both treatment assignment and potential outcomes, with coefficient magnitudes that decrease linearly from 1 to $1/k$, reflecting diminishing covariate relevance. The remaining four covariates serve as irrelevant noise, enabling a sparse confounding structure. This design reflects realistic data environments where only a subset of covariates drive selection and outcomes, while the inclusion of both uniform and normal variables mimics mixed data types commonly observed in empirical applications.

5.2.2 Endogeneity

Endogeneity is induced by drawing the unobserved term v_i in the latent propensity together with the individual errors ε_i from a joint normal distribution in which $\text{Cov}(v_i, \varepsilon_i) = \rho_{v,\varepsilon} \neq 0$. Since v_i influences treatment assignment and is also correlated with the unobserved component of the outcome, the treatment is endogenous. A valid instrument is therefore required to identify causal effects. This construction mirrors empirical settings in which unmeasured factors influence both treatment take-up and response. Full details on the joint-error specification are provided in Appendix E.1.

5.2.3 Instrument

The instrument Z is generated as a binary random variable, $Z \sim \text{Bernoulli}(0.5)$, reflecting settings such as randomized encouragement designs where half the sample receives an external treatment incentive. This binary structure simplifies interpretation and reflects realistic designs in empirical work.

In classical IV settings, instrument strength is often calibrated using the global first-stage F-statistic from a regression of T on (Z, X), with values above $F > 10$ considered "strong". However, a high global first-stage F-statistic does not imply strong local compliance when treatment effects vary with covariates. In such settings, CLATE estimators like GRF and DRIV rely on the conditional shift in treatment probability,

$$\gamma(x) = P(T = 1 \mid X = x, Z = 1) - P(T = 1 \mid X = x, Z = 0),$$

known as the compliance score (Kennedy et al., 2020; Follmann, 2000). While the naive Wald estimator requires only that the average compliance $\mu = \mathbb{E}[\gamma(X)]$ be non-zero, DRIV and GRF demand stronger local identification conditions: $\gamma(x)$ must be bounded away from zero throughout the covariate space. If local compliance is weak, these estimators can become unstable, even when the global F-statistic is large.

This instability is directly linked to the identifying functions of the estimators, as outlined in Section 4. In DMLIV, $\gamma(X)$ determines the conditional variance of the residualised treatment signal, $V_0(X) = \mathrm{Var}[\mathbb{E}[T \mid Z, X] \mid X]$, which must be bounded away from zero for identification. In DRIV, the orthogonal score divides by a local compliance term $\beta(X) = \mathbb{E}[\widetilde{T}\widetilde{Z} \mid X]$, which is proportional to $\gamma(X)$ when $Z \sim \mathrm{Bernoulli}(0.5)$. Lastly, in GRF for IV, $\gamma(X)$ appears in the denominator of each locally weighted regression $\theta(x)$. Thus, all three estimators require some form of local compliance to ensure consistency and valid inference. Appendix C.3 provides full derivations for how each estimator relies on $\gamma(x)$.

To account for this, we consider two instrument calibration regimes in our simulations by altering the value of the parameter π, which governs the extent to which Z shifts the latent treatment score (See Section 5.2.4).

First, we consider a setting with a strong first-stage F-value ($F \approx 15$) but weak compliance. Calibrating π to meet this target yields a low average compliance score ($\mu \approx 0.03$), with many covariate regions exhibiting near-zero compliance. This reflects a situation that would pass the standard first-stage test, but in our setting can still lead to inconsistent estimates. To determine the appropriate value of π, we developed a Monte Carlo search algorithm that targets the desired first-stage F-statistic; see Appendix E.2.1.

Secondly, we consider a setting with a strong compliance score. We tune π to directly target high compliance score, specifically $\mu = 0.5$, ensuring that $\gamma(x)$ is uniformly bounded away from zero. The corresponding value of π is computed using a similar calibration algorithm based on the average compliance score; see Appendix E.2.2. Naturally, this calibration implies a large global first-stage F (on the order of several thousand). However, such a strong compliance score is still realistic in many empirical applications. For example, in the 401(k) dataset discussed in Abadie (2003) and Appendix B, the average compliance score is approximately $\mu = 0.69$, with a corresponding first-stage F-statistic of 12,592.

By contrasting these designs, we assess whether a high F-statistic suffices for local identification. We hypothesize that reliable CLATE estimation requires stronger compliance than what classical first-stage thresholds capture.

5.2.4 Treatment Assignment

Treatment assignment is governed by a continuous latent score that combines observed covariates, the binary instrument and an unobserved error. Specifically, we define

$$d_i \;=\; X_i^\top \beta \;+\; \pi\, Z_i \;+\; v_i,$$

where $X_i^\top \beta$ captures the systematic influence of the first k covariates, π calibrates the instrument's strength and v_i is the unobserved endogenous term. We then convert this latent propensity into a binary treatment indicator by thresholding at its sample median, so that

$$T_i \;=\; \mathbf{1}\bigl\{d_i \geq \text{median}(d)\bigr\}.$$

This thresholding mechanism mirrors empirical contexts, such as eligibility cutoffs based on continuous risk scores or policy thresholds, where an underlying propensity or score must exceed a fixed level to trigger treatment. By using the median, we preserve balance between treated and control units while embedding both confounding and instrument-driven variation directly into the selection rule. Full details on the construction of β are provided in Appendix E.1.

5.2.5 Potential Outcomes

We generate potential outcomes by combining a nonlinear baseline with a heterogeneous treatment effect and correlated noise to mimic endogeneity. Specifically, for each unit i we draw

$$Y_i(0) = g(X_i) + \varepsilon_{i0}, \qquad Y_i(1) = g(X_i) + \theta(X_i) + \varepsilon_{i1},$$

where the smooth baseline is simply

$$g(X_i) = \sin\bigl(X_i^\top \beta\bigr),$$

This sine function introduces mild nonlinearity without dominating the scale of the treatment effect. The disturbances $\varepsilon_{i0}, \varepsilon_{i1}$ are drawn jointly with v_i to induce endogeneity.

We consider two contrasting heterogeneous treatment functions $\theta(X)$. First, A linear heterogeneity, where the treatment effect scales smoothly with a weighted sum of the first k covariates:

$$\theta_{\text{lin}}(X_i) = \zeta\, X_i^\top \beta \;+\; 1,$$

with confounding strength $\zeta = 0.42$ taken from Lechner and Mareckova (2025), reflecting moderate confounding. This linear specification reflects common economic settings where treatment effects vary smoothly with observable characteristics.

Secondly, we consider a stepwise heterogeneity design taken from Wager and Athey (2018), where treatment effects shift abruptly depending on the values of two covariates. Specifically, the treatment function is defined as

$$\theta_{\text{step}}(X_i) = f(X_{i1}) \cdot f(X_{i2}) - 2.8,$$

where $f(x)$ is a modified logistic function that induces a sharp transition near a threshold of one-third. This construction divides the (X_{i1}, X_{i2}) plane into four distinct regions, with a discrete jump in the treatment effect when both covariates exceed the threshold. This setup tests an estimator's ability to detect sudden changes in treatment effects. It mimics policy settings where interventions are triggered only when multiple factors cross specific cutoffs. Full mathematical specifications of $g(X_i)$, $\theta_{\text{lin}}(X_i)$ and $\theta_{\text{step}}(X_i)$, including the form of $f(x)$, can be found in Appendix E.1.

By design, the smooth linear function should pose no difficulty for DRIV, which uses a linear projection to recover smoothly varying effects. In contrast, the stepwise function should play to the strengths of GRF's adaptive tree splits, which can localize estimation in regions where the effect shifts discretely. Comparing performance across these two shapes allows us to see how each method adapts to different forms of heterogeneity under realistic endogeneity.

5.3 Simulation Algorithm and Performance Metrics

5.3.1 Target Estimates and Metrics

For each replication, we recover two target parameters using both the DRIV estimator and the GRF estimator. First, we look at the local average treatment effect (LATE). Then, we compute Individual-level CLATE, i.e. the treatment effect $\theta(x_i)$ for each observation.

Before evaluating estimator performance, we obtain the true values for each target parameter. The population local average treatment effect (LATE) is computed once from an independent large sample of size $N = 10^7$ drawn from the same DGP. Individual treatment effects are calculated exactly from the known functions $\theta(x_i)$ in each test draw.

We compute both the mean and median bias of each estimator across all Monte Carlo replications. The median bias complements the mean by offering a more robust indication of typical performance, as the mean can be distorted by outliers. In addition, we evaluate the mean absolute error (MAE) and root mean squared error (RMSE), which summarize overall estimator accuracy. MAE captures the average magnitude of estimation errors, while RMSE gives greater weight to larger errors, making it particularly informative when assessing the presence of occasional large deviations. Moreover, since RMSE reflects both bias and variance, it is also useful for evaluating the bias–variance trade-off and identifying whether estimation errors are mainly due to systematic bias or variability.

Finally, we compute the 95% confidence interval coverage for the LATE to evaluate the quality of the estimated standard errors. We do not report 95% confidence interval coverage for individual-level treatment effects because several covariates are continuous and each covariate profile X_i occurs at most once across Monte Carlo replications. As a result, the individual-level treatment effect $\theta(X_i)$ is evaluated at essentially unique points in each draw. While we can summarise pointwise accuracy with bias, MAE and RMSE, repeated-sampling coverage for a fixed x is not well-defined. A similar limitation—and practice of focusing on pointwise accuracy—was noted by Lechner and Mareckova (2025).

5.3.2 Simulation Loop and Design Grid

The Monte Carlo simulation is conducted over a fully factorial grid of eight configurations, defined by the cross-product of two sample sizes ($n \in \{1{,}000, 10{,}000\}$), two treatment-effect shapes (linear, step) and two levels of instrument strength: one with high global F-statistic but low compliance (naively strong) and one with high compliance score. In each configuration the correlation between the latent shock and the outcome error is fixed at $\rho_{v,\varepsilon} = 0.5$. The coefficient on the binary instrument, π, is calibrated to match the different instrument strengths using the algorithms in E.2.1 and E.2.2. To ensure reliable performance evaluation across designs, each configuration is simulated using $R = 1{,}000$ replications for $n = 1{,}000$ and $R = 250$ for $n = 10{,}000$, striking a balance between Monte Carlo precision and computational feasibility. The complete simulation workflow is outlined in Algorithm 1.

Algorithm 1 Monte Carlo procedure for DRIV versus GRF

for each simulation grid **do**
  Compute π to target the desired first-stage F–statistic.
  Compute the true Local Average Treatment Effect (LATE) on a large sample of $n = 10{,}000{,}000$.
  Choose number of replications $R \leftarrow 1\,000$ if $n = 1\,000$, else 250.
  for $r = 1, \dots, R$ **do**
    Set $s_{\text{train}} \leftarrow seed_0 + 3\,r$.
    Draw training data $\{X, Z, T, Y, \theta_{\text{true}}\}$ of size n with `seed` $= s_{\text{train}}$.
    Fit the DRIV estimator on $(X, T, Z, Y)_{train}$.
    Fit the IV-Forest estimator on $(X, T, Z, Y)_{train}$.
    Set $s_{\text{test}} \leftarrow s_{\text{train}} + 1$.
    Draw test data of size n with `seed` $= s_{\text{test}}$.
    Predict individual effects $\hat{\theta}_{\text{DRIV}}(X)$ and $\hat{\theta}_{\text{GRF}}(X)$.
    Estimate the Wald Estimator $\hat{\theta}_{\text{Wald}}$
    Compute the true local average treatment effect (LATE).
    Compute LATE metrics: bias, squared error, absolute error, 95% coverage indicator.
    Compute individual effects metrics: mean- and median bias over individuals, squared error and absolute error.
  end for
  Aggregate all replication-level metrics across $r = 1, \dots, R$:
   Mean and median LATE bias, mean absolute error, RMSE and overall coverage rate.
   Mean CLATE bias, RMSE and MAE.
end for

5.3.3 Hyperparameter Settings and Computational Considerations

To ensure comparability and manage computational burden across the simulation grid, all estimators are evaluated using fixed hyperparameters. For the GRF estimator, we follow the default implementation in the `grf` package in R and set the number of trees to 2,000, consistent with Lechner and Mareckova (2025). For the DRIV estimator, all nuisance functions are estimated using random forests configured with `n_estimators` $= 500$ and `max_depth` $= 6$, similar to the DML estimator in Lechner and Mareckova (2025). These settings balance accuracy and computational feasibility, given the scale of the simulation study. While tuning may improve performance, it is beyond the scope of this thesis and would incur prohibitive runtime costs.

6 Results

6.1 Weak Compliance

Table 1: Performance for Conditional LATE and CLATE (n = 1000, weak instrument, linear effect)

	LATE					CLATE			
Estimator	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	−1.2151	−1.1534	2.265	1.8023	1.0000	−1.2167	−1.1575	5.948	4.6715
GRF	−50.5946	0.7308	1186.308	276.6538	0.9640	0.5884	−0.0345	842.592	2.8806
Wald	−0.2780	−0.0251	1.889	0.9934	–	−0.2802	−0.0340	1.946	1.0794

We begin by analyzing estimator performance in settings characterized by weak compliance. When the sample size is small (n = 1,000), both the Doubly Robust IV (DRIV) and Generalized Random Forests (GRF) estimators perform poorly. Table 1 shows the linear treatment effect setting. GRF exhibits an extremely large mean bias in the estimation of the conditional Local Average Treatment Effect (LATE), approximately −50, caused by a small number of extreme outliers. The median bias, however, remains relatively modest at 0.73, which is smaller in magnitude than that of DRIV but still substantially larger than that of the naive estimator. In the estimation of individual-level treatment effects, GRF again suffers from extreme outliers. Although its mean bias is lower than that of DRIV, the root mean squared error (RMSE) exceeds 840, in contrast to values of 5.948 for DRIV and 1.946 for the naive estimator. Interestingly, GRF achieves a lower mean absolute error (MAE) than DRIV for CLATE estimation, suggesting that its high RMSE is driven by a few severe prediction errors. These findings indicate that GRF exhibits higher variance than DRIV under weak compliance, small sample size and linear treatment, despite its lower average bias. Importantly, neither method outperforms the benchmark in any metric.

Table 2: Performance for Conditional LATE and CLATE (n = 1000, weak instrument, step effect)

	LATE					CLATE			
Estimator	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	−0.607	−0.645	4.735	1.240	1.000	−0.680	−0.702	1.458	3.695
GRF	174.644	0.736	3346.69	556.525	0.960	−0.532	−0.062	94.986	2.371
Wald	−0.328	0.003	2.372	1.170	–	−0.403	0.002	2.574	1.459

In the stepwise treatment effect setting, as seen in Table 2, a similar pattern emerges. GRF again suffers from extreme outliers, resulting in an RMSE of 3346.69 for the LATE estimation and 94.986 for the individual-level CLATE estimation. These outliers are most pronounced for the LATE estimate. For the CLATE estimate, the RMSE remains high, but the mean bias, median bias and MAE are lower than that of DRIV. While DRIV produces substantial bias overall, its RMSE remains within an order of magnitude of that of the naive estimator. For CLATE estimation, the RMSE is even lower than that of the naive estimator, although DRIV remains the most biased among the three estimators considered.

Table 3: Performance for Conditional LATE and CLATE (n = 10,000, weak instrument, linear effect)

	LATE					CLATE			
Estimator	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	−1.972	−1.947	2.131	1.972	0.996	−1.973	−1.946	3.126	2.494
GRF	−0.260	0.047	2.836	0.832	0.956	−0.105	−0.096	0.960	0.578
Wald	−0.028	0.003	0.292	0.226	–	−0.030	0.002	0.553	0.444

Increasing the sample size to n = 10,000 decreases the influence of outliers on GRF's performance in the linear treatment setting. As shown in Table 3, GRF produces more stable estimates. Its mean bias in LATE estimation becomes smaller than that of DRIV, although it is still approximately ten times greater in magnitude than that of the naive estimator. For individual-level treatment effects, both RMSE and MAE improve and approach the magnitude of those produced by the naive estimator, although they remain somewhat higher.

Table 4: Performance for Conditional LATE and CLATE (n = 10,000, weak instrument, step effect)

	LATE					CLATE			
Estimator	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	−1.775	−1.769	1.849	1.775	0.992	−1.848	−1.834	2.682	2.167
GRF	112.052	0.095	1773.81	112.911	0.940	−0.173	−0.176	1.076	0.747
Wald	−0.028	−0.008	0.337	0.266	–	−0.102	−0.078	1.033	0.775

In the stepwise treatment function setting shown in Table 4, GRF continues to be affected by outliers. The mean bias for LATE estimation becomes more than 5000 times larger than that of the naive estimator, whereas the median bias is only about 100 times larger. However, GRF becomes competitive with the naive estimator in CLATE estimation. RMSE and MAE are nearly equal across both methods and the mean and median biases of GRF remain modest. Note that this comparison should be interpreted with caution. The naive Wald estimator does not even attempt to model treatment effect heterogeneity. That GRF merely matches its performance, rather than improving upon it still underscores that GRF remains unreliable in settings with weak compliance, despite its apparent improvements.

Although GRF appears to improve in some aspects as sample size increases, DRIV's performance deteriorates when the sample size increases and compliance remains weak. In both the linear and stepwise treatment function settings, the bias increases in magnitude. For the linear treatment function, RMSE and MAE decrease compared to the smaller sample size, especially in the case of CLATE estimation, indicating a reduction in variance. However, for the stepwise treatment function, RMSE increases with the sample size. This increase results partly from larger bias but also suggests that the variance does not decline substantially.

In summary, the results under weak compliance reveal a consistent pattern: neither DRIV nor GRF provides reliable estimates of heterogeneous treatment effects, even as the sample size increases to n = 10,000. GRF shows some stabilization as the sample increases, but remains highly sensitive to outliers, particularly in stepwise treatment functions. DRIV in turn displays persistent and in some cases worsening bias, with limited variance reduction despite the increase in sample size. Across all scenarios with weak instruments, neither method demonstrates an advantage over the naive estimator in scenarios with weak compliance.

6.2 Strong Compliance

Table 5: Performance for Conditional LATE and CLATE (n = 1000, strong instrument, linear effect)

Estimator	**LATE**					**CLATE**			
	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	0.143	0.142	0.237	0.194	1.000	0.143	0.142	0.696	0.548
GRF	0.010	0.012	0.141	0.111	0.953	−0.001	0.000	0.449	0.360
Wald	−0.002	0.003	0.170	0.136	–	−0.002	0.002	0.499	0.403

Table 6: Performance for Conditional LATE and CLATE (n = 1000, strong instrument, step effect)

Estimator	**LATE**					**CLATE**			
	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	−0.323	−0.326	0.371	0.329	1.000	−0.381	−0.383	0.904	0.715
GRF	0.029	0.029	0.148	0.117	0.943	−0.054	−0.051	0.709	0.531
Wald	−0.003	−0.015	0.197	0.158	–	−0.062	−0.074	0.993	0.734

We now turn to the settings with strong instruments. Starting with the linear treatment effect and small sample size (n = 1,000), Table 5 shows that GRF outperforms both DRIV and the naive estimator. It achieves the lowest RMSE and MAE for both LATE and CLATE, along with near-zero bias for CLATE estimation. DRIV, by contrast, performs worst across all metrics. Its bias is an order of magnitude larger than GRF's, and its RMSE and MAE are the highest among the three estimators. Notably, DRIV is not even competitive with the naive estimator for CLATE.

In the stepwise treatment setting (Table 6), GRF's flexibility again proves advantageous. Although it shows slightly higher bias than the naive estimator for LATE, it achieves substantially lower RMSE and MAE. For individual-level estimation, GRF dominates across all metrics. DRIV's performance, in contrast, deteriorates further: it shows the highest bias, and both its RMSE and MAE are barely better than the Wald estimator. Thus, even under strong compliance, DRIV remains unreliable due to persistent bias, while GRF begins to offer clear improvements, especially for CLATE estimation and in the stepwise treatment function. Importantly, GRF now outperforms the naive estimator in CLATE estimation, indicating that it starts to succeed at what it is designed for: capturing treatment effect heterogeneity.

Table 7: Performance for Conditional LATE and CLATE ($n = 10000$, strong instrument, linear effect)

	LATE					CLATE			
Estimator	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	0.123	0.122	0.134	0.123	1.000	0.124	0.122	0.241	0.192
GRF	−0.008	−0.007	0.055	0.040	0.908	−0.011	−0.008	0.346	0.277
Wald	−0.003	0.003	0.055	0.043	–	−0.003	0.002	0.472	0.383

Table 8: Performance for Conditional LATE and CLATE ($n = 10000$, strong instrument, step effect)

	LATE					CLATE			
Estimator	Mean Bias	Median Bias	RMSE	MAE	Coverage Rate	Mean Bias	Median Bias	RMSE	MAE
DRIV	−0.243	0.122	0.499	0.243	0.816	−0.301	0.122	0.641	0.507
GRF	0.045	0.047	0.266	0.059	0.772	−0.022	−0.019	0.212	0.156
Wald	−0.002	−0.001	0.248	0.050	–	−0.061	−0.061	0.975	0.698

Increasing the sample size to $n = 10{,}000$ improves the performance of all estimators in the linear setting (Table 7), but notable differences remain. GRF achieves performance comparable to the Wald estimator for LATE estimation, with similar RMSE but slightly higher bias. DRIV performs substantially worse, with persistent bias and higher overall error. For CLATE, however, GRF is overtaken by DRIV in terms of RMSE and MAE. This suggests that DRIV's parametric projection reduces variance effectively, despite its larger bias. The naive estimator continues to perform well, but we clearly see that the methods targeting treatment effect heterogeneity achieve lower errors under strong compliance.

In the step-effect scenario with large samples (Table 8), GRF again performs best and the differences become most pronounced. For LATE estimation, its RMSE and MAE are nearly identical to those of the naive estimator, though its bias remains slightly larger, implying a lower variance contribution in the RMSE term. For CLATE estimation, GRF clearly dominates: it achieves substantially lower RMSE and MAE than both DRIV and the naive estimator. Specifically, GRF's RMSE is less than one-fifth that of the naive estimator and its MAE is roughly one-quarter. DRIV remains biased in both LATE and CLATE estimation, but for heterogeneous effects it still improves upon the naive method in terms of RMSE and MAE. These results show that GRF is particularly effective in large-sample, nonlinear settings with strong compliance, while DRIV continues to suffer from structural bias even when the sample size is large.

Taken together, these findings show that with large samples and strong compliance, both DRIV and GRF outperform the naive Wald estimator in estimating treatment effect heterogeneity, as reflected in lower RMSE and MAE for CLATE.

6.3 Coverage

Lastly, we examine the coverage of the 95% confidence intervals for the LATE estimates. Across all specifications, DRIV generally produces coverage rates that are too high, indicating excessively conservative confidence intervals. In 5 out of the 8 specifications, the empirical coverage is 100%, including all four cases where $n = 1{,}000$ and one case where $n = 10{,}000$. In the linear treatment setting with a weak instrument and $n = 10{,}000$ seen in Table 3, DRIV achieves a coverage rate of 99.6%, with only one out of 250 replications not including the true LATE. Similarly, in the step function setting with weak compliance and $n = 10{,}000$ in Table 4, the coverage is 99.2%. Notably, the only setting in which DRIV displays undercoverage is in the large sample, strong instrument and step function scenario, where the coverage drops to 81.6%, indicating confidence intervals that are too narrow in this case.

Turning to GRF, the coverage rates are generally close to the nominal level of 95%. In 6 out of 8 cases, the empirical coverage differs from the target level by at most 1.4 percentage points and in 5 of these cases the difference is 1 percentage point or less. However, in the large-sample settings with strong compliance, those where GRF attains the best estimation accuracy, the coverage rates fall below 95%. Table 7 shows that in the linear treatment setting with strong instruments and $n = 10{,}000$, GRF attains a coverage rate of 90.8%. In the stepwise treatment setting under the same conditions, the coverage decreases further to 77.2%. These are also the settings where GRF achieves the best point estimates for LATE and CLATE, suggesting that as the performance of GRF increases, its standard errors may become too narrow.

7 Conclusion and Discussion

This thesis conducted a systematic comparison of two machine learning methods for instrumental variable analysis under endogeneity, namely Doubly Robust IV (DRIV) and Generalized Random Forests (GRF). Using a controlled Monte Carlo simulation framework, it evaluated their ability to estimate heterogeneous treatment effects under realistic conditions that include endogeneity and treatment effect variation. The experimental design varied three dimensions, namely sample size, treatment effect complexity and instrument strength. Estimator performance was assessed across two estimands, the local average treatment effect and individual-level conditional local average treatment effects, using metrics bias, root mean squared error, mean absolute error and confidence interval coverage. The classic Wald estimator was used as a benchmark.

Across all simulation settings, the results reveal substantial differences in the performance of

GRF and DRIV. Under weak instruments, both estimators fail to provide practically meaningful estimates, performing worse than the naive benchmark. GRF suffers from extreme variance due to outliers, while DRIV exhibits persistent and sometimes worsening structural bias. While GRF occasionally achieves lower mean absolute error than the Wald estimator for individual treatment effect estimation, this advantage is undermined by large root mean squared errors, especially in small samples or when treatment effects are nonlinear. Increasing the sample size reduces GRF's variance issues, yet even in large samples, its LATE estimates remain highly sensitive to outliers under weak compliance. Overall, the findings indicate that strong instrument compliance is essential for either method to yield reliable estimates and that under weak compliance, neither GRF nor DRIV offers practically useful inference.

In contrast, strong instruments substantially improve estimator performance. GRF becomes consistently competitive and often superior compared to both DRIV and the Wald estimator, particularly in estimating heterogeneous treatment effects. It delivers low bias, RMSE, and MAE across both linear and stepwise specifications, adapting effectively to treatment effect heterogeneity. GRF is particularly strong when heterogeneity follows a stepwise pattern, as its partitioning structure naturally captures discontinuities and substantially reduces error compared to both the naive IV estimator and DRIV.

Even when the instrument is strong, DRIV continues to exhibit notable bias. One explanation is its reliance on projections. A second possibility is that, in finite samples, the nuisance-function estimates converge more slowly than the $n^{-1/4}$ rate required for orthogonality to eliminate higher-order errors. A third contributing factor may be the use of only two-fold cross-fitting. Interestingly though, DRIV does achieve the best RMSE and MAE in heterogeneous treatment effect estimation under linear conditions when the sample size is large, indicating that its semi-parametric structure can reduce variance when its functional assumptions align with the data-generating process. Apart from linear case with large samples, DRIV is worse than GRF in terms of bias and error.

In general, when compliance is strong, both GRF and DRIV outperform the Wald estimator in estimating heterogeneous treatment effects (CLATE), though DRIV requires a large sample to do so and remains biased, while GRF performs well even in smaller samples. In terms of the LATE, however, the results show that neither DRIV nor GRF consistently outperforms the naive IV estimator, even under favorable conditions.

The coverage analysis highlights further contrasts. For DRIV we see that the standard errors are overly conservative. A plausible explanation is that DRIV's analytic variance formula treats the five high-dimensional nuisance functions as if they were known. When those functions converge more slowly than the rate required by the theory, the extra sampling noise they introduce is ignored, inflating the estimated variance and yielding overly conservative confidence intervals. GRF yields opposite results, with coverage close to the nominal level for most designs. In large-sample and strong-instrument settings, coverage falls below the nominal level. As the point estimates become more stable and sampling variability decreases, the standard errors calculated using bootstrap resampling of fixed-size subsamples fail to capture the remaining variability from random forest weighting and tree construction. This leads to underestimated standard errors and thus under-coverage, precisely when the point estimates are most reliable.

These findings carry important implications. In low-compliance settings, neither GRF nor DRIV can be recommended, as neither method can beat a simple benchmark. However, with strong instruments GRF emerges as the more robust and adaptable method for heterogeneous treatment effect estimation, particularly when treatment effects are nonlinear. DRIV may still offer value in linear contexts with a large sample size, but it comes with substantial bias. If variance is of more concern, then DRIV yields the best results in this case. While both methods offer value when individual-level heterogeneity is of primary interest, they are not generally advisable for researchers interested solely in the average treatment effect for compliers. In such cases, alternative estimators designed specifically for LATE identification may offer greater efficiency and reliability.

These insights are particularly relevant in applied economic settings where treatment effects often vary across individuals and endogeneity arises. In such contexts, misapplying machine learning-based IV methods under weak compliance can result in misleading conclusions about who benefits from an intervention. This risks distorting targeting decisions, inflating or understating program impact, and ultimately leading to inefficient or inequitable policy design. The results thus underscore the importance of aligning estimator choice with the structural features of the empirical setting and the available data, rather than defaulting to complex methods.

While this study offers a structured and informative comparison, several limitations should be acknowledged. First, the analysis is based entirely on simulated data, which cannot capture the full complexity of real-world datasets, such as measurement error or missing observations. The absence of a real-data application means that the external validity of the findings cannot be directly assessed. The design also focuses exclusively on binary treatments and instruments, leaving more general settings with multi-valued treatments and continuous instruments for future work. Moreover, the DRIV estimator relies on projection onto a finite basis, which may introduce bias if the true effect function lies outside the chosen function class. Additionally, hyperparameter tuning was not systematically explored in this study, meaning that the reported estimator performance may not reflect the methods' optimal capabilities and could be sensitive to suboptimal configuration. Finally, the comparison is restricted to two methods chosen for their theoretical grounding and practical accessibility, but the broader landscape of IV-based machine learning estimators remains unexplored.

However, the simulation framework developed in this thesis still offers a flexible and transparent platform for future benchmarking. It can be readily extended to incorporate additional estimators, alternative sources of endogeneity, continuous treatments or instruments and more complex forms of treatment effect heterogeneity, which are natural areas for future research. Applying the framework to empirical datasets also represents a promising area for future research, allowing for validation under real-world data constraints and further enhancing the practical relevance of the comparative findings.

Beyond these immediate extensions, future research could investigate how different model selection and hyperparameter tuning procedures affect estimator performance. Another promising direction lies in further examining the role of compliance strength. The simulations showed that conventional first-stage F-statistics may be insufficient in detecting weak local identification in the presence of heterogeneity. In particular, while methods like DRIV impose a formal lower bound on compliance for identification, they offer no guidance on how to assess or enforce this in practice, a gap that becomes critical when compliance varies across covariates. Future work could explore how these methods behave over a broader spectrum of compliance strength levels and aim to develop a more reliable diagnostic or standardized rule for compliance strength that better aligns with heterogeneous IV estimation, replacing or augmenting the traditional global F-statistic.

Taken together, these findings reinforce the importance of method selection in IV settings with heterogeneous treatment effects. While GRF and DRIV offer valuable tools when used under the right conditions, their limitations, especially under weak compliance, highlight the need for careful diagnostic assessment and further methodological innovation in causal machine learning.

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Appendix

A Code

All code used in this thesis is available in the public GitHub repository: https://github.com/borisrine/Final-thesis-code-613746br. The accompanying README file provides detailed instructions on how to reproduce all results presented in this thesis.

B Replication of Ahrens et al. (2025)

This appendix presents a replication of the simulation study conducted by Ahrens et al. (2025), specifically their Table 2, which is designed to illustrate how different combinations of score construction and cross-fitting affect the finite-sample performance of average treatment effect (ATE) estimators in high-dimensional settings. The purpose of this exercise is to provide an empirical demonstration of the statistical advantages that arise from combining Neyman orthogonality with cross-fitting, the two key components in the double machine learning framework developed by Chernozhukov et al. (2018). While the results in this appendix are not directly related to the core research question of this thesis, they serve to reinforce the broader methodological justification for using debiased machine learning estimators throughout the analysis.

Following the setup described in Appendix B of Ahrens et al. (2025), the simulation is calibrated to real-world data from the 1991 wave of the Survey of Income and Program Participation (SIPP). The outcome variable is net financial assets, the binary treatment is 401(k) eligibility and the covariates include a set of pre-treatment controls. The true ATE in the simulation is derived from a fully nonparametric regression using random forests to estimate the conditional expectations of outcomes and treatment assignment, stratified by treatment status. Using these conditional means, the simulation draws potential outcomes under treatment and control by injecting normal noise with treatment-specific variances. Treatments in the simulation are assigned probabilistically according to the estimated propensity scores. This setup preserves the heterogeneity and confounding structure observed in the original data while allowing repeated sampling for evaluation of estimator performance. For full details on the simulation setup, see Appendix B in Ahrens et al. (2025).

This replication uses 1,000 simulation replications instead of the 10,000 used in Ahrens et al. (2025) for computational feasibility. Despite this reduction, the magnitudes and relative performance patterns closely mirror the findings in Ahrens et al. (2025), confirming the robustness of their conclusions. Random forest models with the same tuning parameters as in the original design are used to estimate the nuisance functions and the simulation framework is implemented in R. The code is included in the supplementary materials of this thesis.

Panel A of Table 9 presents the ATE point estimates and associated standard errors from applying each estimator to the empirical SIPP sample. Panel B displays the key simulation results. The first three estimators (IPW, RA, DR without cross-fitting) rely on plug-in scores and do not use cross-fitting. Among them, IPW (inverse probability weighting) performs poorly in terms of bias and variance, likely due to instability in the estimated propensity scores. RA (regression adjustment) yields smaller bias, but its standard errors are severely underestimated, leading to confidence intervals with poor coverage. Among the three, only the DR estimator uses a Neyman orthogonal score. The doubly robust estimator without cross-fitting performs substantially better than either of the non-orthogonal methods, achieving lower bias and better-calibrated standard errors. This improvement demonstrates the power of orthogonal scores in mitigating the influence of estimation error in nuisance components and thus reducing regularization bias, even in finite samples.

The last three estimators incorporate cross-fitting to address overfitting bias. Cross-fitting by itself improves performance slightly for RA and IPW, but both remain biased and poorly calibrated. In contrast, the combination of cross-fitting with doubly robust scores, i.e. Double Machine Learning, produces the most accurate and reliable estimator in terms of bias and coverage. These results emphasize the complementary nature of Neyman orthogonality and cross-fitting: the former guards against bias from first-stage regularization, while the latter prevents overfitting. Together, they ensure valid inference.

Table 9: Replication of ATE Estimation: Combined Empirical and Simulation Results

	Invalid Estimators					**DML**
	$\check{\theta}_{IPW}$	$\check{\theta}_{RA}$	$\check{\theta}_{DR}$	$\hat{\theta}_{IPW}$	$\hat{\theta}_{RA}$	$\hat{\theta}_{DR}$
Panel A: SIPP Estimates						
Estimate	7505.33	7881.41	7005.13	7510.68	7928.52	6746.33
Std. Error	(1302.38)	(185.19)	(616.68)	(1315.81)	(142.49)	(780.79)
Panel B: Simulation Results						
Mean Bias	-954.20	654.94	142.65	1257.41	628.61	38.76
Median Bias	-928.30	672.40	139.20	1273.30	615.90	40.35
Median Abs. Dev.	1020.33	693.85	429.58	1321.37	651.51	411.64
Std. Deviation	809.93	909.33	1022.93	1709.46	911.45	1052.75
Mean Standard Error	865.72	142.72	556.97	1199.82	131.42	613.99
Coverage Rate (0.95)	0.971	0.234	0.925	0.963	0.212	0.932
Panel C: Essential Components						
Neyman orthogonal	No	No	Yes	No	No	Yes
Cross-fitting	No	No	No	Yes	Yes	Yes

C Proofs and Derivations

C.1 Partial orthogonality of DMLIV

Before we can prove that the DMLIV estimator is only partially orthogonal, we need an intermediary result first, given by the Lemma below.

Lemma 1 (Conditional orthogonality)**.** Under the heterogeneous linear model 3 and the definitions $q_0(X) = \mathbb{E}[Y \mid X],\ p_0(X) = \mathbb{E}[T \mid X],\ h_0(Z, X) = \mathbb{E}[T \mid Z, X]$, we have

$$\mathbb{E}\big[Y - q_0(X) - \theta_0(X)\big(h_0(Z, X) - p_0(X)\big) \;\big|\; Z, X\big] = 0. \tag{9}$$

Proof. Taking $\mathbb{E}[\cdot \mid Z, X]$ in 3 gives $\mathbb{E}[Y \mid Z, X] = \theta_0(X)h_0(Z, X) + f_0(X)$ because $\mathbb{E}[e \mid Z, X] = 0$. Taking $\mathbb{E}[\cdot \mid X]$ in the same equation yields $q_0(X) = \theta_0(X)p_0(X) + f_0(X)$. Thus, we get

$$\begin{aligned}
\mathbb{E}\big[Y - q_0(X) - \theta_0(X)(h_0(Z, X) - p_0(X)) \mid Z, X\big] &= \mathbb{E}[Y \mid Z, X] - q_0(X) - \theta_0(X)(h_0(Z, X) - p_0(X)) \\
&= \big(\theta_0(X)h_0(Z, X) + f_0(X)\big) - \big(\theta_0(X)p_0(X) + f_0(X)\big) - \theta_0(X)(h_0(Z, X) - p_0(X)) \\
&= \theta_0(X)h_0(Z, X) + f_0(X) - \theta_0(X)p_0(X) - f_0(X) - \theta_0(X)h_0(Z, X) + \theta_0(X)p_0(X) \\
&= 0.
\end{aligned}$$

which is exactly 9. □

Proof that DMLIV is orthogonal to q and p, but not to h

Recall that the DMLIV estimator in Stage–0 estimates $\widehat{\theta}_{\text{pre}}(X)$ by regressing the residualised outcome $\widetilde{Y}$ on the residualised treatment $\widetilde{T}$, where

$$\widetilde{Y} = Y - \widehat{q}(X), \qquad \widetilde{T} = \widehat{h}(Z, X) - \widehat{p}(X).$$

Regressing $\widetilde{Y}$ on $\widetilde{T}$ is, in population, equivalent to choosing $\theta(\cdot)$ that minimises the conditional squared-error loss

$$\ell(\theta) \;=\; \mathbb{E}\big[(\widetilde{Y} - \theta(X)\,\widetilde{T})^2 \mid X\big].$$

Taking the derivative of $\ell(\theta)$ with respect to $\theta(X)$ gives the first-order condition

$$\mathbb{E}\left[(Y - q(X) - \theta(X)(h(Z, X) - p(X)))(h(Z, X) - p(X)) \mid X\right] = 0.$$

Hence, for any candidate pair (θ, q, p, h) we can write the score

$$m^{\text{DMLIV}}\big(X; \theta, q, p, h\big) \;=\; -2\,\mathbb{E}\big[(Y - q(X) - \theta(X)(h(Z, X) - p(X)))(h(Z, X) - p(X)) \mid X\big].$$

The Stage-0 estimator plugs in the learners $(\hat{q}, \hat{p}, \hat{h})$ and solves

$$m^{\text{DMLIV}}(X; \theta, \hat{q}, \hat{p}, \hat{h}) = 0.$$

Step 1: orthogonality in q and p.

To show that this score is orthogonal to q and p, we evaluate the classical derivatives of m^{DMLIV} with respect to q and p at the true nuisance functions (q_0, p_0, h_0) and the true parameter θ_0:

$$\begin{aligned}
\nabla_{q(X)}\, m^{\text{DMLIV}}\big(X; \theta_0, q_0, p_0, h_0\big) &= -2\,\mathbb{E}\big[h_0(Z, X) - p_0(X) \mid X\big] = 0, \\
\nabla_{p(X)}\, m^{\text{DMLIV}}\big(X; \theta_0, q_0, p_0, h_0\big) &= -2\,\theta_0(X)\,\mathbb{E}\big[h_0(Z, X) - p_0(X) \mid X\big] \\
&\quad + 2\,\mathbb{E}\big[Y - q_0(X) - \theta_0(X)\{h_0(Z, X) - p_0(X)\} \mid X\big] = 0.
\end{aligned}$$

Both expectations are zero by the conditional moment restrictions used to identify θ_0. Hence small errors in $\widehat{q}$ or $\widehat{p}$ affect the score only in second order—exactly the Neyman-orthogonality property we want.

Step 2: failure of orthogonality in h.

Because the first-stage function h depends on the instrument Z, the usual partial derivative with respect to $h(X)$ is not defined. This is because h is a function of the pair (Z, X), while $m^{\text{DMLIV}}(\cdot, \cdot \mid X)$ conditions only on X. In other words, inside $\mathbb{E}[\cdot \mid X]$ the instrument Z is still random, so the score depends on the full surface $h(Z, X)$. A usual partial derivative with respect to $h(X)$ is therefore undefined.

We therefore inspect the directional derivative in the direction $\nu(Z, X) = h(Z, X) - h_0(Z, X)$, where $h_0(Z, X)$ is the true first–stage regression of the treatment on the instrument and covariates. This yields

$$
\begin{aligned}
D_h\, m^{\text{DMLIV}}\big(X; \theta_0, q_0, p_0, h_0\big)[\nu] := {} & 2\,\theta_0(X)\big[(h_0(Z, X) - p_0(X))\,\nu(Z, X) \mid X\big] \\
& + 2\big[(Y - q_0(X) - \theta_0(X)\{h_0(Z, X) - p_0(X)\})\,\nu(Z, X) \mid X\big] \\
= {} & 2\,\theta_0(X)\big[(h_0(Z, X) - p_0(X))\,\nu(Z, X) \mid X\big],
\end{aligned}
$$

where the second term equals 0 by Lemma 1.

The final quantity is not necessarily zero, since

$$
\mathbb{E}\left[h_0(Z, X) - p_0(X) \mid Z, X\right] \neq 0.
$$

This is reasonable because we are using $h(Z, X)$ as part of our regressor; hence any error in the measurement of that regressor propagates directly to an error in $\theta(X)$. The derivative would have been zero only if the residual error $h(Z, X) - h_0(Z, X)$ were independent of the residual randomness $h_0(Z, X) - p_0(X)$ conditional on X. In general the two are correlated: the second term measures how far the true first-stage $h_0(Z, X)$ deviates from its conditional mean $p_0(X) = \mathbb{E}[h_0(Z, X) \mid X]$, while the first term measures how far the estimate $h(Z, X)$ deviates from $h_0(Z, X)$. It is therefore highly probable that the values of Z that lead to a large deviation from the mean treatment are precisely those for which the first-stage model makes larger mistakes.

□

C.2 Proof that the DRIV estimator is fully orthogonal

We show that the expected derivative of the moment (the derivative of the loss with respect to $\theta(X)$) conditional on X, with respect to each of the nuisance functions, is equal to zero when the true nuisance and target functions are plugged in.

Let the nuisance vector be $g(X) = \big(\theta_{\text{pre}}(X),\, p(X),\, q(X),\, r(X),\, \beta(X)\big)$, where, under the true data-generating process,

$$p_0(X) = [T \mid X], \quad q_0(X) = [Y \mid X], \quad r_0(X) = [Z \mid X], \quad \beta_0(X) = \mathbb{E}[(T - p_0(X))(Z - r_0(X)) \mid X].$$

and

$$f_0(x) \;=\; q_0(x) \;-\; \theta_0(x)\, p_0(x).$$

For any (θ, g) define the DRIV moment

$$m^{DRIV}\big(X; \theta(X), g(X)\big) \;=\; \mathbb{E}\Big[\theta_{\text{pre}}(X) + \frac{\big(Y - q(X) - \theta_{\text{pre}}(X)\,(T - p(X))\big)\,(Z - r(X))}{\beta(X)} - \theta(X) \;\Big|\; X\Big].$$

For a direction $\nu = \theta' - \theta$, the directional derivative of the loss with respect to θ is

$$-\,2\,\mathbb{E}\big[m^{DRIV}\big(X; \theta(X), g(X)\big)\,\nu(X)\big].$$

Therefore, to establish Neyman-orthogonality it is enough to show that the (classical) derivative of m^{DRIV} with respect to every component of $g(X)$ vanishes at the truth (θ_0, g_0).

Evaluating each gradient component at (θ_0, g_0):

$$\begin{aligned}
\nabla_{\theta_{\text{pre}}(X)}\, m^{DRIV}(X; \theta_0(X), g_0(X)) &:= \;\mathbb{E}\left[1 - \frac{(T - p_0(X))(Z - r_0(X))}{\beta_0(X)} \;\middle|\; X\right] \\
&= \;1 - \frac{\mathbb{E}[(T - p_0(X))(Z - r_0(X)) \mid X]}{\beta_0(X)} = 1 - \frac{\beta_0(X)}{\beta_0(X)} \;=\; 0,
\end{aligned}$$

$$\begin{aligned}
\nabla_{p(X)}\, m^{DRIV}(X; \theta_0(X), g_0(X)) &:= \;\theta_0(X)\,\mathbb{E}\Big[\frac{Z - r_0(X)}{\beta_0(X)} \;\Big|\; X\Big] \;=\; \theta_0(X)\,\left[\mathbb{E}\Big[\frac{Z}{\beta_0(X)} \;\Big|\; X\Big] - \mathbb{E}\Big[\frac{r_0(X)}{\beta_0(X)} \;\Big|\; X\Big]\right] \\
&= \;\theta_0(X)\left[\mathbb{E}\Big[\frac{Z}{\beta_0(X)} \;\Big|\; X\Big] - \mathbb{E}\Big[\frac{Z}{\beta_0(X)} \;\Big|\; X\Big]\right] = 0,
\end{aligned}$$

$$\nabla_{q(X)}\, m^{DRIV}(X; \theta_0(X), g_0(X)) := \;-\,\mathbb{E}\Big[\frac{Z - r_0(X)}{\beta_0(X)} \;\Big|\; X\Big] \;=\; 0,$$

$$\begin{aligned}
\nabla_{r(X)}\, m^{DRIV}(X; \theta_0(X), g_0(X)) &:= \;-\,\mathbb{E}\Big[\tfrac{Y - q_0(X)}{\beta_0(X)} \;\Big|\; X\Big] + \theta_0(X)\,\mathbb{E}\Big[\tfrac{T - p_0(X)}{\beta_0(X)} \;\Big|\; X\Big] \\
&= \;-\,\frac{1}{\beta_0(X)}\big(\mathbb{E}[Y \mid X] - q_0(X)\big) + \frac{\theta_0(X)}{\beta_0(X)}\big(\mathbb{E}[T \mid X] - p_0(X)\big) \\
&= \;-\,\frac{1}{\beta_0(X)}\big(q_0(X) - q_0(X)\big) + \frac{\theta_0(X)}{\beta_0(X)}\big(p_0(X) - p_0(X)\big) \\
&= \;0,
\end{aligned}$$

$$\begin{aligned}
\nabla_{\beta(X)}\, m^{DRIV}(X; \theta_0(X), g_0(X)) &:= \;-\,\mathbb{E}\Big[\frac{\big(Y - q_0(X) - \theta_0(X)(T - p_0(X))\big)\,(Z - r_0(X))}{\beta_0(X)^2} \;\Big|\; X\Big] \\
&= \;-\,\mathbb{E}\Big[\frac{\mathbb{E}\big[Y - q_0(X) - \theta_0(X)(T - p_0(X)) \mid Z, X\big]\,(Z - r_0(X))}{\beta_0(X)^2} \;\Big|\; X\Big]
\end{aligned}$$

$$= -\frac{\mathbb{E}\Big[\mathbb{E}\big[Y - \theta_0(X)\,T - f_0(X)\;\big|\;Z, X\big]\,(Z - r_0(X))\;\Big|\;X\Big]}{\beta_0(X)^2}$$

$$= -\frac{1}{\beta_0(X)^2}\,\mathbb{E}\Big[(Z - r_0(X))\;\underbrace{\mathbb{E}\big[Y - q_0(X) - \theta_0(X)(T - p_0(X))\;\big|\;Z, X\big]}_{=\;\mathbb{E}[\varepsilon\,\mid\,Z,X]=0}\;\Big|\;X\Big]$$

$$= -\frac{1}{\beta_0(X)^2}\,\mathbb{E}\big[\,0\cdot(Z - r_0(X))\mid X\big]\;=\;0.$$

Hence, the DRIV moment is orthogonal to all components of $g(X)$ and is thus Neyman Orthogonal.

□

C.3 Derivations for reliance on compliance scores

This appendix shows how the local compliance score

$$\gamma(X)\;=\;\Pr\big(T=1\,\big|\,X, Z=1\big)\;-\;\Pr\big(T=1\,\big|\,X, Z=0\big) \tag{A.1}$$

controls the key denominators and variances that enter the three estimators studied in Section 4. Throughout the derivation the instrument satisfies $Z \sim \text{Bernoulli}(0.5)$.

First-stage representation

Let

$$h_0(Z,X)\;=\;\mathbb{E}\big[T\,\big|\,Z,X\big],\qquad p_0(X)\;=\;\mathbb{E}\big[T\,\big|\,X\big].$$

Because the instrument is binary with $\Pr(Z=1)=\Pr(Z=0)=\frac{1}{2}$, denote the two conditional means

$$h_0(1,X)=\mathbb{E}[T\mid Z=1,X],\qquad h_0(0,X)=\mathbb{E}[T\mid Z=0,X],$$

and define the local compliance score

$$\gamma(X)\;=\;h_0(1,X)\;-\;h_0(0,X).$$

Any binary variable can be written as $Z=\mathbb{1}\{\text{treatment encouraged}\}$. Hence,

$$h_0(Z,X)\;=\;h_0(0,X)\;+\;\big(h_0(1,X)-h_0(0,X)\big)\,Z\;=\;h_0(0,X)\;+\;\gamma(X)\,Z. \tag{A.2}$$

Next, average **??** over Z conditional on X to obtain the unconditional first stage:

$$\begin{aligned} p_0(X)\;&=\;\mathbb{E}\big[T\,\big|\,X\big]=\mathbb{E}\big[h_0(Z,X)\,\big|\,X\big]\\ &=h_0(0,X)\;+\;\gamma(X)\,\mathbb{E}[Z\mid X]\\ &=h_0(0,X)\;+\;\tfrac{1}{2}\,\gamma(X), \end{aligned} \tag{A.3}$$

because $\mathbb{E}[Z\mid X]=\frac{1}{2}$.

Finally, the conditional mean of the instrument is

$$r_0(X) \;=\; \mathbb{E}\big[Z \,\big|\, X\big] \;=\; \tfrac{1}{2}. \tag{A.4}$$

The population versions of the residualised treatment and instrument defined in Section 4 are

$$\widetilde{T} = h_0(Z,X) - p_0(X) = \gamma(X)\left(Z - \tfrac{1}{2}\right). \tag{A.5}$$

$$\widetilde{Z} = Z - r_0(X) = Z - \tfrac{1}{2}. \tag{A.6}$$

DMLIV: conditional variance $V_0(X)$

Equation 4 in the main text defines

$$V_0(X) \;=\; \mathrm{Var}\big(\mathbb{E}[T \mid Z, X] \,\big|\, X\big) \;=\; \mathrm{Var}\big(\widetilde{T} \,\big|\, X\big).$$

Using A.5,

$$V_0(X) \;=\; \gamma^2(X)\ \mathrm{Var}\big(Z - \tfrac{1}{2}\big) \;=\; \gamma^2(X)\ \mathrm{Var}(Z) \;=\; \gamma^2(X) \times \tfrac{1}{4}. \tag{A.7}$$

DRIV: local compliance term $\beta_0(X)$

Section 4.1.2 defines

$$\beta_0(X) \;=\; \mathbb{E}\big[\widetilde{T}\,\widetilde{Z} \,\big|\, X\big].$$

Substituting A.5–A.6 yields

$$\beta_0(X) \;=\; \gamma(X)\,\mathbb{E}\big[(Z - \tfrac{1}{2})^2\big] \;=\; \gamma(X) \times \tfrac{1}{4}. \tag{A.8}$$

IV–GRF: local Wald denominator

For IV estimation in GRF, the locally weighted IV regression uses

$$\mathrm{Cov}(T, Z \mid X) \;=\; \mathbb{E}\big[\big(T - p_0(X)\big)\left(Z - \tfrac{1}{2}\right) \big|\, X\big] \;=\; \mathbb{E}\big[\widetilde{T}\,\widetilde{Z} \,\big|\, X\big] \;=\; \beta_0(X) \;=\; \gamma(X) \times \tfrac{1}{4}. \tag{A.9}$$

Implications

Equations A.7 to A.9 show that when $Z \sim \mathrm{Bernoulli}(0.5)$, all three estimators depend on the compliance score. DMLIV requires that $V_0(X) \geq \lambda > 0$, which is equivalent to $\gamma(X)^2 \geq 4\lambda$. DRIV requires $\beta_0(X) \geq \beta_{\min} > 0$, which implies $\gamma(X) \geq 4\beta_{\min}$. IV–GRF also requires that the conditional covariance in A.9 does not vanish, leading to the same condition. As a result, if $\gamma(X)$ becomes close to zero in any region of the covariate space, each estimator experiences high variance or numerical instability, regardless of the global first-stage F-statistic. Calibrating the data-generating process based on the average compliance score $\mu = \mathbb{E}[\gamma(X)]$ directly targets the identification conditions that ensure reliable estimation in finite samples with heterogeneity.

D Technical details of standard error and confidence interval estimation in generalized Random Forests

In this appendix we summarize the key results fromAthey et al. (2019) needed to construct standard errors and Wald-style confidence intervals for the forest estimate $\hat{\theta}_n(x)$.

D.1 Asymptotic normality

Under Assumptions 1–6 of Athey et al. (2019) and with honest subsampling trees, theorem 5 of their paper shows that the generalized random forest estimate $\hat{\theta}_n(x)$ satisfies

$$\frac{\hat{\theta}_n(x) - \theta(x)}{\sigma_n(x)} \xrightarrow{d} \mathcal{N}(0, 1).$$

Combining Theorem 5 with the influence-function representation in their Eq. (15) gives the asymptotic variance

$$\sigma_n^2(x) \;=\; \xi^\top \Big[V(x)^{-1}\, H_n\big(x;\, \theta(x), \eta(x)\big)\, V(x)^{-1\top} \Big]\, \xi.$$

Here:

- ξ is the unit vector selecting the θ–coordinate from (θ, η).
- $$V(x) \;=\; \frac{\partial}{\partial(\theta, \eta)}\, \mathbb{E}\big[\psi_{\theta,\eta}(W) \;\big|\; X = x\big]\bigg|_{(\theta,\eta)=(\theta(x),\eta(x))},$$

  the Jacobian ("curvature") of the population moment function.
- $$H_n\big(x; \theta, \eta\big) \;=\; \mathrm{Var}\Big(\sum_{i=1}^{n} \alpha_i(x)\, \psi_{\theta,\eta}(W_i)\Big),$$

  the variance of the forest-weighted scores at (θ, η), with weights $\alpha_i(x)$ as in (3) of the main text.

D.2 Estimation of $V(x)$

The curvature matrix $V(x)$ can be estimated by a plug-in rule. For the IV GRF setting, this becomes

$$V(x) = \begin{pmatrix} \mathbb{E}[Z_i T_i \mid X_i = x] & \mathbb{E}[Z_i \mid X_i = x] \\ \mathbb{E}[T_i \mid X_i = x] & 1 \end{pmatrix},$$

and each conditional expectation may be estimated via an honest regression forest, evaluated at x. Denote the resulting estimate by $\widehat{V}_n(x)$.

D.3 The bootstrap-of-little-bags for $H_n(x)$

Following Sexton and Laake (2009) and Section 4.1 of Athey et al. (2019), we estimate $H_n\big(x; \hat{\theta}_n(x), \hat{\eta}_n(x)\big)$ by grouping the B trees into G "little bags" of size ℓ each ($B = G\ell$). Write

$$\hat{\Psi}_b(x) \;=\; \sum_{i=1}^{n} \alpha_{b,i}(x)\, \psi_{\hat{\theta}_n(x), \hat{\eta}_n(x)}(W_i) \quad (b = 1, \ldots, B)$$

for the score from tree b and for each bag $m = 1, \ldots, G$ let

$$\hat{\Psi}_m(x) \;=\; \frac{1}{\ell} \sum_{b=(m-1)\ell+1}^{m\ell} \hat{\Psi}_b(x), \qquad \bar{\Psi}(x) \;=\; \frac{1}{G} \sum_{m=1}^{G} \hat{\Psi}_m(x).$$

Define the "between-bag" and "within-bag" mean-squared deviations:

$$B(x) = \frac{1}{G-1} \sum_{m=1}^{G} \big(\hat{\Psi}_m(x) - \bar{\Psi}(x)\big)^2,$$

$$W(x) = \frac{1}{G\,(\ell-1)} \sum_{m=1}^{G} \sum_{b=(m-1)\ell+1}^{m\ell} \big(\hat{\Psi}_b(x) - \hat{\Psi}_m(x)\big)^2.$$

Then the bootstrap-of-little-bags estimator for the score variance is

$$\widehat{H}_n(x) \;=\; B(x) \;-\; \frac{W(x)}{\ell}.$$

D.4 Final variance estimator and confidence intervals

Combining these pieces, we form the plug-in variance estimator

$$\widehat{\sigma}_n^2(x) \;=\; \xi^\top \Big[\widehat{V}_n(x)^{-1}\, \widehat{H}_n(x)\, \widehat{V}_n(x)^{-1\top}\Big]\, \xi, \quad \widehat{\mathrm{SE}}(x) = \sqrt{\widehat{\sigma}_n^2(x)}.$$

An approximate Wald confidence interval for $\theta(x)$ is then

$$\big[\,\hat{\theta}_n(x) \;\pm\; z_{1-\alpha/2}\,\widehat{\mathrm{SE}}(x)\big],$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$–quantile of the standard normal.

E Simulation Design Details

E.1 Data-Generating Utilities

In this appendix we collect the precise definitions of the functions used to generate all components of our simulated data. Although our implementation follows the structure of Lechner and Mareckova (2025) Appendix B, we extend it by introducing a binary instrument and a flexible procedure to calibrate its strength.

Covariate generation.

We simulate $p = u + n$ covariates for each unit $i = 1, \ldots, N$, partitioned into u "uniform" and n "normal" features. Specifically,

$$X_i = \left(X_i^U, X_i^N\right), \quad X_i^U \sim \mathcal{U}\left(-\tfrac{\sqrt{12}}{2}, \tfrac{\sqrt{12}}{2}\right)^u, \quad X_i^N \sim \mathcal{N}(0,1)^n.$$

By construction $\mathrm{Var}(X_{ij}^U) = \mathrm{Var}(X_{ij}^N) = 1$, so all covariates enter with unit variance. We choose $u = n = p/2$ and fix $p = 10$. However, the design allows for any value of p, u and n. The Python and R implementation also allow the user to change these parameters.

Error term generation.

To induce endogeneity we draw a 3-dimensional disturbance vector $(v_i, \varepsilon_{i0}, \varepsilon_{i1})$ jointly from

$$\begin{pmatrix} v_i \\ \varepsilon_{i0} \\ \varepsilon_{i1} \end{pmatrix} \sim \mathcal{N}\Big(0, \Sigma\Big), \qquad \Sigma = \begin{pmatrix} 1 & \rho_{v\varepsilon} & \rho_{v\varepsilon} \\ \rho_{v\varepsilon} & 1 & 0 \\ \rho_{v\varepsilon} & 0 & 1 \end{pmatrix},$$

so that $\mathrm{Cov}(v_i, \varepsilon_{i,d}) = \rho_{v,\varepsilon}$ and $\mathrm{Cov}(\varepsilon_{i0}, \varepsilon_{i1}) = 0$. This structure embeds a latent confounder v_i that affects both treatment assignment and outcomes. We fix $\rho_{v,\varepsilon} = 0.5$. However, this can be changed within the Python and R implementation by the user.

Instrument.

We draw a binary instrument

$$Z_i \sim \mathrm{Bernoulli}\big(0.5\big),$$

independently across i. This mimics randomized encouragement designs and ensures $\mathbb{E}[Z_i] = 0.5$, while satisfying the exclusion restriction by construction.

Latent index and treatment assignment.

Define a sparse coefficient vector

$$\beta = \left(1, 1 - \tfrac{1}{k}, 1 - \tfrac{2}{k}, \ldots, 1 - \tfrac{k-1}{k}, 0, \ldots, 0\right)^\top \in R^p,$$

where only the first k covariates (with $k = 6$ in our thesis) drive selection and outcome. We normalize by

$$\psi = \sqrt{\sum_{j=1}^{p} \beta_j^2 / 1.25}$$

to match the scaling in Lechner and Mareckova (2025).

Remark 1 (Why divide by 1.25?). To mirror the simulation design of Lechner and Mareckova (2025), we rescale the observable part of the latent index,

$$\eta_i \;:=\; \frac{X_i^\top \beta}{\psi}$$

Because every covariate is centred and standardised, $\mathbb{E}[X_{ij}] = 0$ and $\mathrm{Var}(X_{ij}) = 1$, it follows that

$$\mathrm{Var}\big(X_i^\top \beta\big) = \sum_{j=1}^{p} \beta_j^2, \quad \Longrightarrow \quad \mathrm{Var}(\eta_i) = \frac{\sum_{j=1}^{p} \beta_j^2}{\sum_{j=1}^{p} \beta_j^2 / 1.25} = 1.25.$$

Fixing $\mathrm{Var}(\eta_i) = 1.25$ achieves two goals:

- Comparability across designs. When p or the number of active coefficients k changes, the raw scale of $X_i^\top \beta$ would vary wildly; the normalisation anchors it to a common metric.
- Stable non-linear transformations. The link functions used by Lechner and Mareckova (2025) (logistic, sine, quadratic, step) exhibit enough curvature for $\mathrm{sd}(\eta_i) \approx 1.12$; a smaller scale would make them almost linear, whereas a much larger scale would push them into their flat tails. In our thesis, this is relevant for the step function.

The instrument Z_i enters additively in the latent propensity score

$$d_i \;=\; \frac{X_i^\top \beta}{\psi} \;+\; \pi\, Z_i \;+\; v_i,$$

with π governing instrument relevance and v_i capturing idiosyncratic noise. It follows that

$$\mathrm{Var}(d_i) = \; 1.25 \;+\; \pi^2\, \mathrm{Var}(Z_i) \;+\; \mathrm{Var}(v_i),$$

so the strength of selection stemming from the covariates is held constant, while we can still tune instrument strength via π.

We assign treatment by thresholding at the sample median of $\{d_i\}_{i=1}^{N}$:

$$T_i = \mathbf{1}\big\{d_i \geq \mathrm{median}_j(d_j)\big\},$$

which yields a balanced treated–control split and preserves the analog of an eligibility cutoff.

Baseline function

We follow Lechner and Mareckova (2025) in combining a smooth baseline with deliberately chosen heterogeneous treatment-effect functions that stress different features of the competing estimators.

The baseline component

$$g(X_i) = \delta\, \sin\!\Big(\tfrac{X_i^\top \beta}{\psi}\Big), \qquad \delta = 1,$$

injects mild non-linearity while preserving a zero mean and unit variance for the systematic part of the outcome.

Linear heterogeneity.

The first treatment function varies smoothly with the same affine index that drives selection:

$$\tau_{\text{lin}}(X_i) \;=\; \frac{\zeta\, X_i^{\top}\beta}{\psi} + 1, \quad \zeta = 0.42.$$

We set $\zeta = 0.42$ to match the medium-strong confounding as in Lechner and Mareckova (2025). This parameter can be changed to get different levels of confounding by the covariates.

Stepwise heterogeneity.

The second specification produces abrupt changes in the treatment effect that are hard to approximate with global polynomials but are well captured by adaptive forests. Let

$$f^{WA}(x) = 1 + \frac{1}{1 + \exp\bigl\{-20\,(x - \frac{1}{3})\bigr\}},$$

$$\tau_{\text{step}}(X_i) = f^{WA}(x_{i1})\, f^{WA}(x_{i2}) - 2.8.$$

The logistic transform $f^{WA}(\cdot)$, introduced by Wager and Athey (2018) and used by Lechner and Mareckova (2025), is nearly flat at 1 for $x < \frac{1}{3}$ and rises steeply to 2 for $x > \frac{1}{3}$. Multiplying the two terms yields four regions in the (x_{i1}, x_{i2}) plane, with a large jump in τ_{step} once both covariates exceed the threshold. The constant -2.8 centres the population average treatment effect around 1, ensuring comparability with the linear case.

Intuitively, this design mimics policy rules that activate only when multiple risk factors cross pre-set cut-offs. Estimators such as GRF, which recursively partition the covariate space, should localise quickly on the high-effect quadrant, whereas DRIV must rely on higher-order interactions in its first-stage regressions. By contrasting τ_{lin} and τ_{step} we therefore obtain a clean test of smooth-versus-sudden heterogeneity under otherwise identical endogeneity.

Potential outcomes

Finally, potential outcomes are assembled as

$$Y_i(0) = g(X_i) + \varepsilon_{i0}, \qquad Y_i(1) = g(X_i) + \tau(X_i) + \varepsilon_{i1},$$

and the observed outcome obeys the usual equation $Y_i = (1 - T_i)\, Y_i(0) + T_i\, Y_i(1)$.

E.2 Algorithms to calibrate π

E.2.1 Calibration using the first-stage F

To generate a naive "strong" instrument with first-stage F-statistic around 15, we calibrate the instrument strength parameter π via a data-driven search procedure. Specifically, we solve for the value of π that yields a small-sample first-stage F from a regression of T on Z and X close to the target $F = 15$ on average. This ensures that instrument strength satisfies conventional first stage-IV thresholds while preserving the complex data-generating structure described above.

The procedure proceeds as follows:

1. **Simulate and compute $F(\pi)$ for a single draw:**
   - Generate a dataset of size N, drawing covariates X, binary instrument $Z \sim \text{Bernoulli}(0.5)$ and correlated errors v, ε.
   - Construct latent treatment scores $d_i = X_i^\top \beta + \pi Z_i + v_i$ and assign treatment as $T_i = \mathbf{1}\{d_i \geq \text{median}(d)\}$.
   - Run the OLS regression:
     $$T_i = \alpha + \gamma Z_i + X_i^\top \eta + \nu_i,$$
     and extract the F-statistic $F(\pi)$ testing $\gamma = 0$.
2. **Compute the average first-stage statistic over Monte Carlo replications:**
   - Repeat Step 1 independently $M = 500$ times with varying random seeds.
   - Compute the Monte Carlo average:
     $$\overline{F}(\pi) = \frac{1}{M} \sum_{m=1}^{M} F^{(m)}(\pi).$$
3. **Solve for π using the bisection method:**
   - Initialize the search bracket: $\pi_{\text{low}} = 0.01$, $\pi_{\text{high}} = 10$.
   - Evaluate $\overline{F}(\pi_{\text{low}})$ and $\overline{F}(\pi_{\text{high}})$ to ensure that $F_{\text{target}} = 15$ lies within the bracket.
   - For up to 500 iterations, update:
     $$\begin{aligned} \pi_{\text{mid}} &= \frac{1}{2}(\pi_{\text{low}} + \pi_{\text{high}}), \\ \text{If } \overline{F}(\pi_{\text{mid}}) < 15, &\text{ set } \pi_{\text{low}} = \pi_{\text{mid}}, \\ \text{Else, set } \pi_{\text{high}} &= \pi_{\text{mid}}. \end{aligned}$$
   - Stop when $|\overline{F}(\pi_{\text{mid}}) - 15| < 10^{-3}$ or after 500 iterations.
4. **Output calibrated π values:** This routine is applied twice—once for $N = 1{,}000$ and once for $N = 10{,}000$—to account for sample-size effects on the F-statistic. The resulting values π_{1000} and π_{10000} are then used in the main simulations to ensure a conventionally strong first stage across both designs.

All in all, this yields the following Algorithm 2.

Algorithm 2 Monte Carlo Calibration of π via First-Stage F-Statistic

Input: Target F-statistic F_{target}, sample size N, correlation $\rho_{v,\varepsilon}$, number of repetitions $M = 500$, tolerance $\epsilon = 10^{-3}$, maximum iterations $= 500$

Initialize search bracket: $\pi_{\text{low}} \leftarrow 0.01$, $\pi_{\text{high}} \leftarrow 10$

Check bracketing condition: Ensure $\overline{F}(\pi_{\text{low}}) < F_{\text{target}} < \overline{F}(\pi_{\text{high}})$

for iteration $= 1$ to 500 **do**

  $\pi_{\text{mid}} \leftarrow \frac{1}{2}(\pi_{\text{low}} + \pi_{\text{high}})$

  For $m = 1, \ldots, M$ **do**

  - Simulate dataset with covariates X, instrument Z and errors (v, ε)
  - Compute $d_i = X_i^\top \beta + \pi_{\text{mid}} Z_i + v_i$, assign $T_i = \mathbf{1}\{d_i \geq \text{median}(d)\}$
  - Run OLS: $T_i \sim Z_i + X_i$ and extract $F^{(m)}(\pi_{\text{mid}})$ testing Z

  Compute average $\overline{F}(\pi_{\text{mid}}) = \frac{1}{M} \sum_{m=1}^{M} F^{(m)}(\pi_{\text{mid}})$

  if $|\overline{F}(\pi_{\text{mid}}) - F_{\text{target}}| < \epsilon$ **then**

    return π_{mid}

  else if $\overline{F}(\pi_{\text{mid}}) < F_{\text{target}}$ **then**

    $\pi_{\text{low}} \leftarrow \pi_{\text{mid}}$

  else

    $\pi_{\text{high}} \leftarrow \pi_{\text{mid}}$

  end if

end for

return final midpoint $\pi = \frac{1}{2}(\pi_{\text{low}} + \pi_{\text{high}})$

This algorithm is implemented in `piFinderF.py` and available at: https://github.com/borisrine/Final-thesis-code-613746br.

E.2.2 Calibration by average compliance score

To calibrate the instrument so that the average compliance score

$$\mu(\pi) = \mathbb{E}\big[P(T = 1 \mid X, Z = 1) - P(T = 1 \mid X, Z = 0)\big]$$

hits a prespecified target μ_{target} (e.g. $\mu_{\text{target}} = 0.5$), we solve for π via the following bisection procedure:

1. **Compute one-draw compliance score $\widehat{\mu}(\pi)$:**
   - Generate a dataset of size N: covariates X, $Z \sim \text{Bernoulli}(0.5)$ and errors $(v, \varepsilon_0, \varepsilon_1)$.
   - Form $d_i = X_i^\top \beta + \pi\, Z_i + v_i$ and assign $T_i = \mathbf{1}\{d_i \geq \text{median}(d)\}$.
   - Compute
   $$\widehat{\mu}(\pi) = \mathbb{E}[T_i \mid Z_i = 1] - \mathbb{E}[T_i \mid Z_i = 0].$$
2. **Monte Carlo averaging:**
   - Repeat Step 1 independently $M = 500$ times with distinct random seeds.

- Compute the average

$$\overline{\mu}(\pi) \;=\; \frac{1}{M}\sum_{m=1}^{M}\widehat{\mu}^{(m)}(\pi).$$

3. **Bisection search for π:**

   - Initialize $\pi_{\text{low}} = 0.01,\ \pi_{\text{high}} = 10$ and verify $\overline{\mu}(\pi_{\text{low}}) < \mu_{\text{target}} < \overline{\mu}(\pi_{\text{high}})$.
   - For up to 500 iterations:

$$\pi_{\text{mid}} = \tfrac{1}{2}(\pi_{\text{low}} + \pi_{\text{high}}), \quad \mu_{\text{mid}} = \overline{\mu}(\pi_{\text{mid}}).$$

     Update the bracket:

$$\begin{cases} \pi_{\text{low}} = \pi_{\text{mid}}, & \text{if } \mu_{\text{mid}} < \mu_{\text{target}}, \\ \pi_{\text{high}} = \pi_{\text{mid}}, & \text{otherwise.} \end{cases}$$

     Stop early if $\left|\mu_{\text{mid}} - \mu_{\text{target}}\right| < 10^{-3}$.

4. **Output:** The solution $\pi_{\text{compliance}}$ such that $\overline{\mu}\left(\pi_{\text{compliance}}\right) \approx \mu_{\text{target}}$.

All in all, this yields the following Algorithm 3.

Algorithm 3 Monte Carlo Calibration of π via Compliance Score

Input: Target compliance μ_{target}, sample size N, correlation $\rho_{v,\varepsilon}$, repetitions $M = 500$, tolerance $\epsilon = 10^{-3}$, max iterations $= 500$
Initialize search bracket: $\pi_{\text{low}} \leftarrow 0.01$, $\pi_{\text{high}} \leftarrow 10$
Check bracketing: Ensure $\overline{\mu}(\pi_{\text{low}}) < \mu_{\text{target}} < \overline{\mu}(\pi_{\text{high}})$
for iteration $= 1$ to 500 **do**
  $\pi_{\text{mid}} \leftarrow \frac{1}{2}(\pi_{\text{low}} + \pi_{\text{high}})$
  For $m = 1, \ldots, M$ **do**
  - Simulate data (X, Z, T) using $\pi = \pi_{\text{mid}}$
  - Compute compliance score $\widehat{\mu}^{(m)} = \mathbb{E}[T \mid Z = 1] - \mathbb{E}[T \mid Z = 0]$

  Compute average compliance: $\overline{\mu}(\pi_{\text{mid}}) = \frac{1}{M}\sum_{m=1}^{M}\widehat{\mu}^{(m)}$
  if $|\overline{\mu}(\pi_{\text{mid}}) - \mu_{\text{target}}| < \epsilon$ **then**
    return π_{mid}
  else if $\overline{\mu}(\pi_{\text{mid}}) < \mu_{\text{target}}$ **then**
    $\pi_{\text{low}} \leftarrow \pi_{\text{mid}}$
  else
    $\pi_{\text{high}} \leftarrow \pi_{\text{mid}}$
  end if
end for
return final midpoint $\pi = \frac{1}{2}(\pi_{\text{low}} + \pi_{\text{high}})$

This routine is implemented in `piFinderCompliance.py` in the repository: github.com/borisrine/Final-thesis-code-613746br.